T2K 1.431×10^{20}-POT *3-Flavour* Muon-Neutrino Disappearance Analysis

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Abstract

We present the results of a 3-flavour ν_μ-disappearance analysis on the combined T2K Run 1+2 dataset, which corresponds to an integrated J-PARC neutrino beam exposure of 1.431×10^{20} POT. This analysis used 3-flavour oscillation probabilities instead of the 2-flavour approximation that was used in our earlier analysis reported in Ref. [1]. Also oscillated ν_e templates were included in this analysis in the interaction modes CCQE, CC1π and CC other, whereas these were neglected in the 2-flavour analysis.

In the 3-flavour analysis, the total predicted number of single μ-like ring events in Super-K is higher for the inverted hierarchy (IH) than for the normal hierarchy (NH) if the atmospheric mass-squared difference is $|\Delta m^2_{32}|$. This results from the prediction of more events above the oscillation maximum for the IH than for the NH and less below it, which is due to the 3-flavour ν_μ-survival probability being higher for the IH than the NH above the oscillation maximum and lower below the oscillation maximum. Since there are more events above the oscillation maximum than below it, the total predicted number is higher for the IH than for the NH. This is true for any combination of the non-23 oscillation parameters.

Uncertainties in δ_{CP} can change the predicted reconstructed spectrum of single μ-like ring events in Super-K by up to 5% above the oscillation maximum and by up to 3% below it. Meanwhile changes in the spectrum due to uncertainties in $\sin^2(2\theta_{13})$, $\sin^2(2\theta_{12})$ and Δm^2_{12} are relatively small at $\leq 0.5\%$.

A 3-flavour ν_μ-disappearance fit was made of the observed Super-K reconstructed energy spectrum of single μ-like ring events in the combined Run 1+2 dataset. In this fit, $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ were varied to find their best-fit values, but the non-23 oscillation parameters were fixed. Using GENIE, this gave best-fit values of $\sin^2(2\theta_{23}) = 0.98$, $|\Delta m^2_{32}| = 2.6 \times 10^{-3}$ eV^2/c^4 for both mass hierarchies, while a fit using NEUT gave $\sin^2(2\theta_{23}) = 0.99$, $|\Delta m^2_{32}| = 2.6 \times 10^{-3}$ eV^2/c^4 for the NH and $\sin^2(2\theta_{23}) = 0.99$, $|\Delta m^2_{32}| = 2.7 \times 10^{-3}$ eV^2/c^4 for the IH. For comparison, the best-fit values from the 2-flavour analysis in [1] were $\sin^2(2\theta_{23}) = 0.97$, $|\Delta m^2_{32}| = 2.6 \times 10^{-3}$ eV^2/c^4 using GENIE and $\sin^2(2\theta_{23}) = 0.98$, $|\Delta m^2_{32}| = 2.6 \times 10^{-3}$ eV^2/c^4 using NEUT. The 3-flavour best-fit points were higher in $\sin^2(2\theta_{23})$ than those in the 2-flavour analysis due to the effect of $\sin^2(2\theta_{13})$: the 3-flavour analysis used the value $\sin^2(2\theta_{13}) = 0.1$, whereas the 2-flavour analysis used the approximation $\sin^2(2\theta_{13}) = 0$.

Uncertainties in the best-fit points due to uncertainties in δ_{CP} were larger than those due to uncertainties in $\sin^2(2\theta_{13})$, $\sin^2(2\theta_{12})$ and Δm^2_{12}; these uncertainties due to δ_{CP} were 0.5% (NH) and 0.35% (IH) in $\sin^2(2\theta_{23})$ and 0.6% (NH) and 0.4% (IH) in $|\Delta m^2_{32}|$. However this is relatively small compared with the combined systematic uncertainties in the best-fit points which were 4.3 - 5.1%.

The 3-flavour confidence contours are narrower in $\sin^2(2\theta_{23})$ than the 2-flavour contours since the 3-flavour best-fit points are nearer to $\sin^2(2\theta_{23}) = 1.0$. The confidence contours for the IH include higher values of $|\Delta m^2_{32}|$ than those for the NH; this is because it is necessary to choose a higher value of this parameter for the IH than for the NH in order to fit the oscillation dip at a certain energy in the real data.

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1. Introduction

In this note we present the results of a 3-flavour ν_μ-disappearance analysis on the combined T2K Run1+2 dataset. This work builds on the effective 2-flavour ν_μ-disappearance analysis of the Run1+2 dataset that was reported in Ref. [1]. Further technical details of the effective 2-flavour analysis may be found in the analysis of the Run 1 dataset that was reported in Ref. [2].

As the measurement error on the atmospheric Δm^2 approaches the solar Δm^2 ($\sim$0.075$\times10^{-3}$ eV^2/c^4), subleading effects of the solar Δm^2 on ν_μ-disappearance analyses should not be neglected. This was pointed out by E.Lisi in the XIV International Workshop on Neutrino Telescopes [5], and brought to the attention of the T2K ν_μ-disappearance contour committee by M. Mezzetto. In response, the contour committee recommended moving to a 3-flavour analysis [6].

The method for predicting the nominal Super-K reconstructed energy spectrum of single μ-like ring events is described in Ref. [1] and it is unchanged. This work considers the same systematic uncertainties as Ref. [1]. As in the effective 2-flavour analysis, the best-fit oscillation parameters were found using the likelihood-ratio method [14], and the confidence regions were constructed using the Feldman-Cousins method [16]. Systematic errors were included in the oscillation contours using the Cousins-Highland method (Bayesian integration) and by determining their effects on the Feldman-Cousins $\Delta\chi^2_{critical}$ values.

The oscillation probabilities in this work were calculated in a full 3-flavour framework including matter effects. The ν_μ-survival probability depends on sin^{2}2θ_{12}, sin^{2}2θ_{13}, Δm^2_{12}, δ_{CP} and the neutrino mass hierarchy as well as sin^{2}2θ_{23} and $|\Delta m^2_{32}|$. Throughout this note $\sin^2(2\theta_{12})$, $\sin^2(2\theta_{13})$, Δm^2_{21}, δ_{CP} will be collectively denoted the "non-23" oscillation parameters. The method of calculation of the 3-flavour oscillation probabilities is described in Sec. 2, and details of the validation of the code and our assessment of the its numerical accuracy are also presented. The predicted reconstructed single μ-like ring spectra in Super-K for the 3-flavour analysis are shown in Sec. 3. Details of the 3-flavour fits of toy MC datasets are given in Sec. 4; in this analysis, sin^{2}2θ_{12}, sin^{2}2θ_{13}, Δm^2_{21} and δ_{CP} were randomised when creating the toy MC datasets but were fixed when fitting them or the real data. This section also includes the results of the 3-flavour fit of the Run1+2 dataset for both GENIE and NEUT, and a comparison with the results of the 2-flavour analysis [1]. The 3-flavour confidence regions are shown in Sec. 5, and these are compared with those obtained in the 2-flavour analysis [1] and with previous results from MINOS [17] and Super-K [18]. A summary is given in Sec. 6.

2. 3-flavour neutrino oscillation probabilities

In this analysis, 3-flavour neutrino oscillation probabilities in constant density matter are used. Vacuum 3-flavour oscillation probabilites are also calculated, but these are used only to estimate the accuracy of the 3-flavour probabilities in matter (see section 2.6). The method of calculation of the vacuum probabilities is given in section 2.1, while the calculation of the probabilities in constant density matter is described in section 2.2. The code that makes these calculations has been integrated into the Rutherford/Oxford/Lancaster oscillation analysis package. In section 2.4, comparisons are given between 2-flavour and 3-flavour (in matter) oscillation probabilities, and between 3-flavour oscillation probabilities in vacuum and matter. Validation of the code is described in section 2.5: the vacuum probabilities were validated by comparing their values with those obtained from an alternative formulation in the PDG review. Several validation checks of the probabilities in constant density matter are made as they are calculated, and the values of the probabilities were checked by comparing them with those given by an independently-written program. Finally, an estimate of the accuracy of the oscillation probabilities in matter is given in section 2.6.

2.1. 3-flavour neutrino oscillation probabilities in vacuum

The 3-flavour neutrino oscillation probabilities in vacuum are calculated from equation 11 in [9]:

$$P(\nu_\alpha \rightarrow \nu_\beta) = \delta_{\alpha\beta} - 4\sum_{i>j} Re(U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j})\sin^2\left(\frac{\Delta m^2_{ij}L}{4E}\right) + 2\sum_{i>j} Im(U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j})\sin\left(\frac{\Delta m^2_{ij}L}{2E}\right) \quad (1)$$

In equation 1, Δm^2_{ij} = m^2_j - m^2_i, L is the distance from the neutrino target in Tokai to Super-K (295 km), E is the neutrino energy, and U is the PMNS matrix in vacuum:

$$U = \begin{pmatrix} c_{12}c_{13} & c_{13}s_{12} & s_{13}e^{-i\delta} \\ -c_{23}s_{12} - s_{13}s_{23}c_{12}e^{i\delta} & c_{12}c_{23} - s_{12}s_{13}s_{23}e^{i\delta} & c_{13}s_{23} \\ s_{12}s_{23} - s_{13}c_{12}c_{23}e^{i\delta} & -s_{23}c_{12} - s_{12}c_{23}s_{13}e^{i\delta} & c_{13}c_{23} \end{pmatrix} \quad (2)$$

with $s_{12} = \sin(\theta_{12})$, $c_{12} = \cos(\theta_{12})$, etc. $P(\bar{\nu}_\alpha \rightarrow \bar{\nu}_\beta)$ is calculated by changing the sign of the third term in equation 1 (this assumes that CPT is conserved).

2.2. 3-flavour neutrino oscillation probabilities in constant density matter

All three neutrino flavours undergo neutral-current interactions with protons, neutrons and electrons in matter. These neutral-current interactions have identical amplitudes for all three flavours, and they produce no observable effects on the probabilities of oscillation between one flavour and another. However the oscillation probabilities are changed by coherent forward scattering of electron neutrinos in charged-current interactions with electrons in matter [9].

Using natural units with $\hbar$ = c = 1, the time evolution of neutrino flavour states in vacuum is described by the Schrödinger equation:

$$i\frac{\partial\psi}{\partial t} = H_F\psi \tag{3}$$

where

$$\psi = \begin{pmatrix} \psi_e \\ \psi_\mu \\ \psi_\tau \end{pmatrix} \tag{4}$$

and H_F is the effective Hamiltonian in the flavour-state basis

$$H_F = \frac{1}{2E}UMU^\dagger \tag{5}$$

with M defined by

$$M = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \Delta m^2_{21} & 0 \\ 0 & 0 & \Delta m^2_{31} \end{pmatrix} \tag{6}$$

The calculation of oscillation probabilities requires neutrino flight time to be converted to distance travelled, and this conversion assumes that neutrinos are highly relativistic. In this calculation, $UMU^\dagger$ is used for neutrinos, but its complex conjugate is used for antineutrinos. For neutrinos, the effects of the charged-current interactions between electron neutrinos and electrons are taken into account by adding the potential $2E\sqrt{2}G_FN_e$ to the real part of the first diagonal element of $UMU^\dagger$, where E is the neutrino energy, G_F is the Fermi coupling constant, and N_e is the number density of electrons in matter. For antineutrinos, this potential is subtracted from the real part of the first diagonal element of the complex conjugate of $UMU^\dagger$ (for further details see section 9.2 in [10]).

After the addition or subtraction of this potential, $UMU^\dagger$ (or its complex conjugate) is diagonalised. The eigenvalues of a Hermitian matrix are always real; they are calculated by solving the (cubic) characteristic equation using the method of del Ferro, Tartaglia and Cardano as described in [12]. The differences between the eigenvalues are the effective mass-squared differences in matter. The eigenvectors are calculated using an algebraic method: one of the components is set to be 1.0 (real), and the other two components are calculated using two of the three simultaneous equations in $UMU^\dagger - \lambda I = 0$, where the λ are the eigenvalues. After being normalised, the three eigenvectors become the columns of the effective mixing matrix in matter.

The initial flavour state of the neutrino is represented by a 1×3 column vector whose entries are complex; if, for example, the initial flavour state is a muon neutrino, this column vector is

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

The vector representing the initial flavour state is then multiplied by the Hermitian conjugate of the matter mixing matrix to convert it to another 1×3 complex column vector representing the initial mass states. These mass states are propagated to Super-K by multiplying the jth initial mass state by $\exp\left(\frac{-i\Delta m^2_{j1}L}{2E}\right)$, where the Δm^2_{j1} are the mass-squared differences in matter, i.e. the differences between the eigenvalues of $UMU^\dagger$. This gives a new 1×3 complex column vector representing the final mass states, and these are converted to the final flavour states by multiplying by the matter mixing matrix. The entries of the resulting 1×3 complex column vector represent the amplitudes of each flavour at Super-K, and the corresponding oscillation probabilities are calculated as the moduli squared of these amplitudes.

2.3. Inputs to the 3-flavour neutrino oscillation probability calculations

When calculating the 3-flavour oscillation probabilities, the mixing angles can be input either in the form $\sin^2(2\theta_{ij})$ or as $\sin^2(\theta_{ij})$. $\sin(\theta_{ij})$ and $\cos(\theta_{ij})$ are calculated from the input values, and these are used in turn to calculate the elements

of U. The "solar" mass-squared difference is input as Δm^2_{21}, and this is > 0 for both mass hierarchies. The "atmospheric" mass-squared difference is input either as $|\Delta m^2_{32}|$ or using a definition introduced by G. Fogli, E. Lisi et al. in [8]:

$$\Delta m^2_{FL} = m^2_3 - \frac{m^2_2 + m^2_1}{2} \tag{7}$$

The quantity denoted Δm^2_{FL} is useful since its absolute value is the same for both mass hierarchies, and only its sign changes. If the atmospheric mass-squared difference is input as Δm^2_{32}, the mass hierarchy is set using the sign of Δm^2_{32}; this is positive for the normal mass hierarchy (NH) and negative for the inverted mass hierarchy (IH). In this case, Δm^2_{31} is not input but is calculated as $\Delta m^2_{31} = \Delta m^2_{21} + \Delta m^2_{32}$ for both hierarchies. If the atmospheric mass-squared difference is input as Δm^2_{FL}, the mass hierarchy is set using the sign of Δm^2_{FL}. Then Δm^2_{32} and Δm^2_{31} are calculated from the input values of Δm^2_{FL} and Δm^2_{21}.

2.4. Results of the 3-flavour neutrino oscillation probability calculations

A comparison of 2-flavour and 3-flavour (in matter) ν_μ-survival probabilities is given in figure 1. In this figure, the nominal probabilities are calculated with $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4, δ_{CP} = 0 and Earth crust density = 2.6 g/cm^3. The bands represent the maximum and minimum 3-flavour probabilities from 4000 random variations of the non-23 oscillation parameters within their uncertainties at each energy with the same values of $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ being used. The uncertainties in $\sin^2(2\theta_{12})$ and Δm^2_{21} are taken from KamLAND combined with solar experiments [19], while those in $\sin^2(2\theta_{13})$ are computed as a weighted average of measurements from Daya Bay [20] and Reno [21].

The effect of $\sin^2(2\theta_{13})$ on the 3-flavour ν_μ-survival probability in matter is shown in figure 2: there is a small increase in P($\nu_\mu \rightarrow \nu_\mu$) at all energies as $\sin^2(2\theta_{13})$ increases. Also shown in figure 3 is the effect of δ_{CP} on the 3-flavour ν_μ-survival probability in matter: as δ_{CP} increases from 0 to π, P($\nu_\mu \rightarrow \nu_\mu$) increases above the oscillation maximum and decreases below it for the NH, with these changes being reversed for the IH. The effects of uncertainties in $\sin^2(2\theta_{12})$ and Δm^2_{21} on the 3-flavour P($\nu_\mu \rightarrow \nu_\mu$) in matter are shown for the NH in table 1. At 0.6 GeV, uncertainties in $\sin^2(2\theta_{12})$ change P($\nu_\mu \rightarrow \nu_\mu$) by $\approx$ 0.00025 and uncertainties in Δm^2_{21} change it by $\approx$ 0.0001.

For the NH, the 3-flavour P($\nu_\mu \rightarrow \nu_\mu$) is increased below the oscillation maximum and decreased above it compared with the 2-flavour probability, while for the IH these changes are reversed. The oscillation maximum is at lower energy for the IH than for the NH, and this can be understood more easily by considering the vacuum probabilities. In equation 1, the imaginary terms are zero if $\alpha = \beta$, and only the 3 real terms contribute to the vacuum P($\nu_\mu \rightarrow \nu_\mu$). If the atmospheric mass-squared difference is input as Δm^2_{32}, the terms $|U_{\mu 1}|^2$, $|U_{\mu 2}|^2$, $|U_{\mu 3}|^2$, $sin^2\left(\frac{\Delta m^2_{21} L}{4E}\right)$ and $\sin^2\left(\frac{\Delta m^2_{32} L}{4E}\right)$ are identical for the two hierarchies, and the only term that changes between them is $\sin^2\left(\frac{\Delta m^2_{31} L}{4E}\right)$. Since Δm^2_{31} is lower for the IH than for the NH, $\sin^2\left(\frac{\Delta m^2_{31} L}{4E}\right)$ has its maximum at a lower energy for the IH than for the NH. The vacuum P($\nu_\mu \rightarrow \nu_\mu$) depends on 1 - 4 $|U_{\mu 1}|^2|U_{\mu 3}|^2$ $\sin^2\left(\frac{\Delta m^2_{31} L}{4E}\right)$, which means that its minimum is at a lower energy for the IH than for the NH.

Referring to the bands in figure 1, the highest ν_μ-survival probabilities for the NH are for low values of $\sin^2(2\theta_{12})$, high values of $\sin^2(2\theta_{13})$ and δ_{CP} near π above the oscillation maximum, while below the maximum they are for high values of $\sin^2(2\theta_{12})$ and $\sin^2(2\theta_{13})$ and δ_{CP} near 0. For the IH, the highest probabilities are for high values of $\sin^2(2\theta_{12})$ and $\sin^2(2\theta_{13})$ and δ_{CP} near 0 above the oscillation maximum, but for low values of $\sin^2(2\theta_{12})$, high values of $\sin^2(2\theta_{13})$ and δ_{CP} near π below it.

Comparisons of ν_μ oscillation probabilities in vacuum and matter are given for the NH in table 2 and for the IH in table 3. At 0.6 GeV, P($\nu_\mu \rightarrow \nu_e$) is increased by $\approx$ 0.004 and P($\nu_\mu \rightarrow \nu_\tau$) is decreased by $\approx$ 0.004 in matter compared with vacuum for the NH, while for the IH the changes are once more reversed. The ν_μ-survival probability is relatively unaffected by the passage of ν_μ through matter, and is changed by only $\approx$ 0.0001 compared with the vacuum probability at 0.6 GeV.

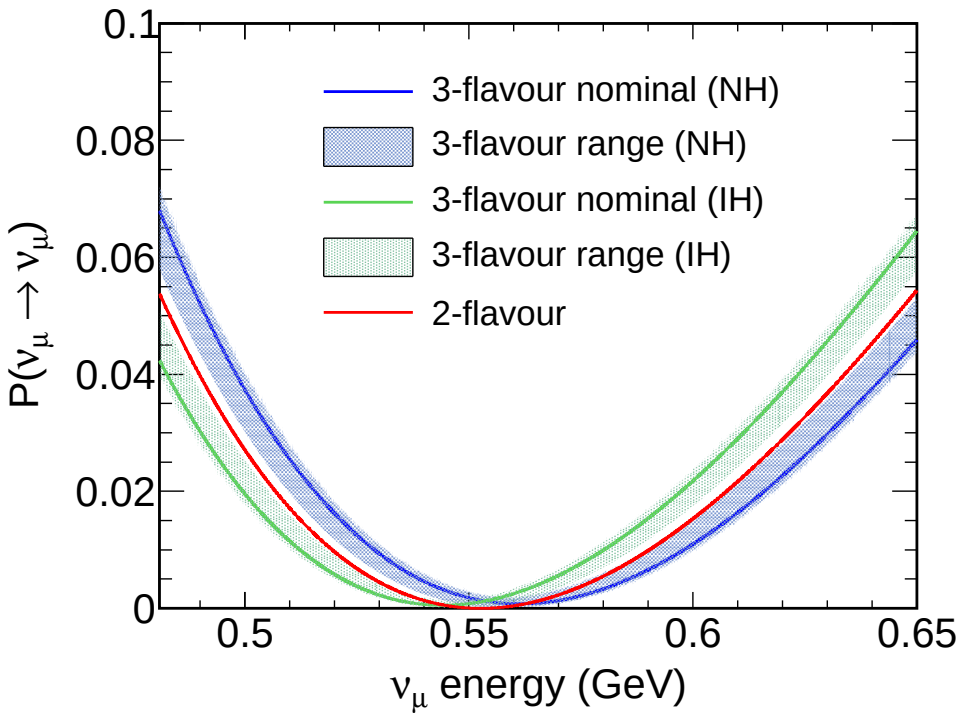


Figure 1: Comparison of P($\nu_\mu \rightarrow \nu_\mu$) for 2-flavour and 3-flavour (in matter) oscillations, NH = normal mass hierarchy, IH = inverted mass hierarchy. The nominal probabilities are calculated with $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4, δ_{CP} = 0 and Earth crust density = 2.6 g/cm^3. The bands represent the maximum and minimum 3-flavour probabilities from 4000 random variations of the non-23 oscillation parameters within their uncertainties at each energy with the same values of $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ being used.

Table 1: Effects of uncertainties in $\sin^2(2\theta_{12})$ and Δm^2_{21} on the 3-flavour P($\nu_\mu \rightarrow \nu_\mu$) in matter; this is for the normal mass hierarchy, $\sin^2(2\theta_{23})$ = 1.0, $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4, $\sin^2(2\theta_{13})$ = 0.1 and δ_{CP} = 0

Energy (GeV)	P($\nu_\mu \rightarrow \nu_\mu$) for $\sin^2(2\theta_{12})$ = 0.870, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4	P($\nu_\mu \rightarrow \nu_\mu$) for $\sin^2(2\theta_{12})$ = 0.906, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4	P($\nu_\mu \rightarrow \nu_\mu$) for $\sin^2(2\theta_{12})$ = 0.833, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4	P($\nu_\mu \rightarrow \nu_\mu$) for $\sin^2(2\theta_{12})$ = 0.870, Δm^2_{21} = 7.8 $\times 10^{-5}$ eV^2/c^4	P($\nu_\mu \rightarrow \nu_\mu$) for $\sin^2(2\theta_{12})$ = 0.870, Δm^2_{21} = 7.4 $\times 10^{-5}$ eV^2/c^4
0.3	0.95703	0.95809	0.95608	0.95733	0.95674
0.4	0.35053	0.35247	0.34877	0.35135	0.34971
0.5	0.037395	0.038009	0.036842	0.037663	0.037129
0.6	0.010963	0.010692	0.011212	0.010841	0.011086
0.7	0.094387	0.093711	0.095004	0.094083	0.094691
0.8	0.20453	0.20371	0.20528	0.20416	0.20490
0.9	0.31099	0.31015	0.31175	0.31061	0.31137
1.0	0.40463	0.40383	0.40535	0.40427	0.40499
1.1	0.48414	0.48340	0.48481	0.48380	0.48447
1.2	0.55080	0.55013	0.55141	0.55050	0.55111
1.3	0.60656	0.60594	0.60711	0.60628	0.60683
1.4	0.65329	0.65274	0.65380	0.65304	0.65354
1.5	0.69265	0.69215	0.69311	0.69243	0.69288
1.6	0.72599	0.72554	0.72640	0.72579	0.72620
1.7	0.75440	0.75399	0.75478	0.75421	0.75459
1.8	0.77876	0.77839	0.77910	0.77859	0.77893
1.9	0.79978	0.79943	0.80009	0.79962	0.79993
2.0	0.81801	0.81770	0.81830	0.81787	0.81815

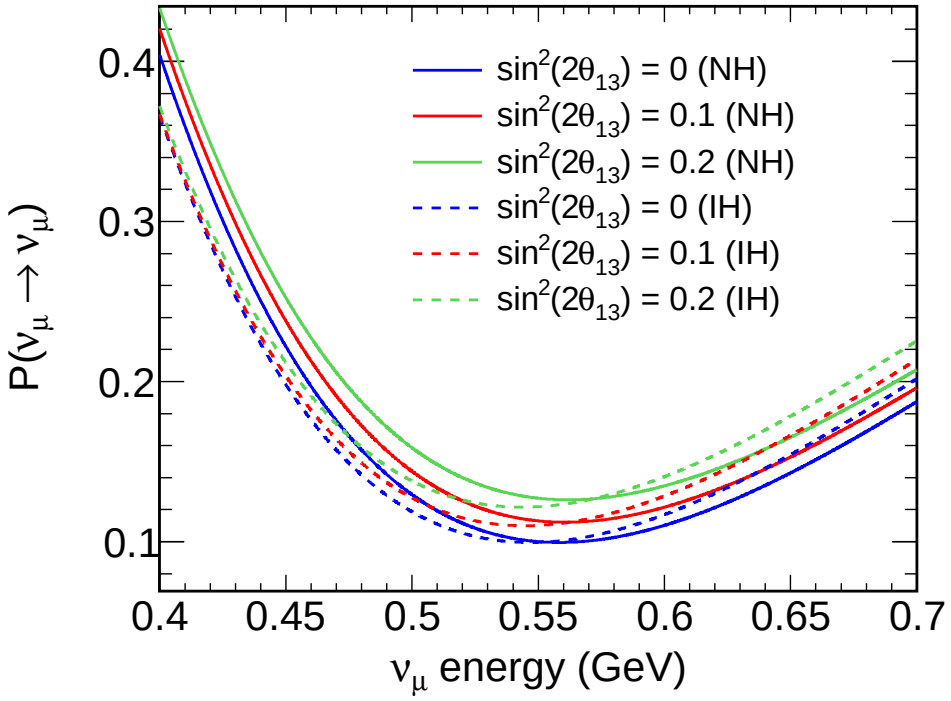


Figure 2: Effect of $\sin^2(2\theta_{13})$ on P($\nu_\mu \rightarrow \nu_\mu$) for 3-flavour oscillations in matter; NH = normal mass hierarchy, IH = inverted mass hierarchy, $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4, δ_{CP} = 0 and Earth crust density = 2.6 g/cm^3

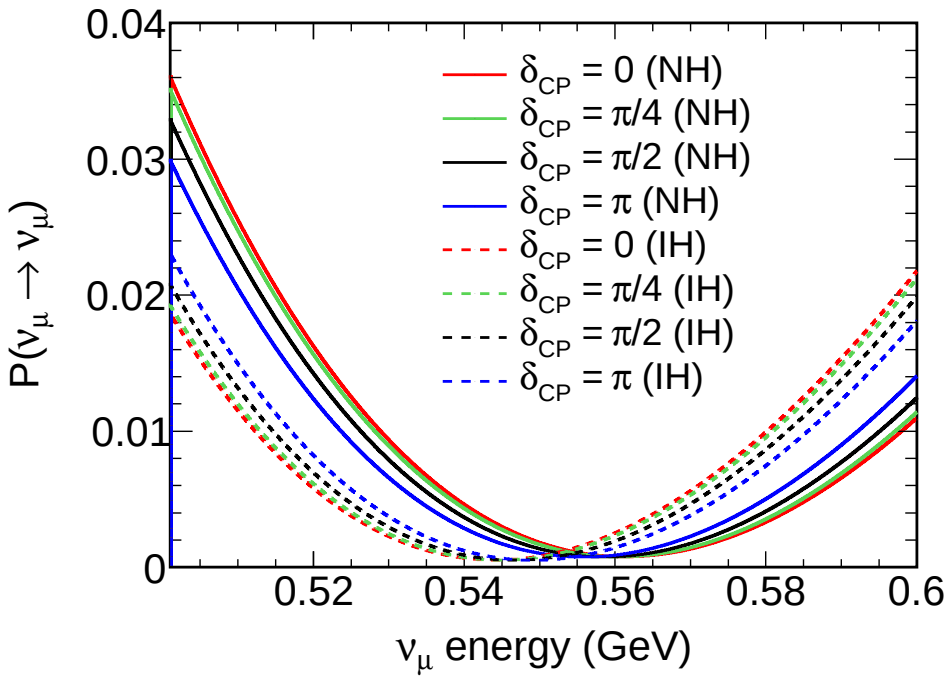


Figure 3: Effect of δ_{CP} on P($\nu_\mu \rightarrow \nu_\mu$) for 3-flavour oscillations in matter; NH = normal mass hierarchy, IH = inverted mass hierarchy, $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4 and Earth crust density = 2.6 g/cm^3

Table 2: Comparison of ν_μ oscillation probabilities in vacuum and matter for the normal mass hierarchy, Earth crust density = 2.6 g/cm^3, $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times$ 10^{-5} eV^2/c^4, $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4 and δ_{CP} = 0

Energy (GeV)	P($\nu_\mu \rightarrow \nu_e$)		P($\nu_\mu \rightarrow \nu_\mu$)		P($\nu_\mu \rightarrow \nu_\tau$)	
	Vacuum	Matter	Vacuum	Matter	Vacuum	Matter
0.3	0.00054403	0.00016981	0.95496	0.95703	0.044501	0.042796
0.4	0.024253	0.029051	0.34954	0.35053	0.62621	0.62042
0.5	0.046291	0.051402	0.037071	0.037395	0.91664	0.91120
0.6	0.051975	0.056108	0.010883	0.010963	0.93714	0.93293
0.7	0.049739	0.052871	0.094385	0.094387	0.85588	0.85274
0.8	0.044830	0.047181	0.20455	0.20453	0.75062	0.74829
0.9	0.039478	0.041256	0.31101	0.31099	0.64951	0.64776
1.0	0.034502	0.035865	0.40464	0.40463	0.56085	0.55951
1.1	0.030139	0.031200	0.48415	0.48414	0.48571	0.48466
1.2	0.026403	0.027241	0.55081	0.55080	0.42279	0.42196
1.3	0.023233	0.023905	0.60656	0.60656	0.37021	0.36954
1.4	0.020548	0.021093	0.65329	0.65329	0.32616	0.32562
1.5	0.018268	0.018715	0.69266	0.69266	0.28908	0.28863
1.6	0.016325	0.016696	0.72599	0.72599	0.25769	0.25731
1.7	0.014661	0.014971	0.75440	0.75440	0.23094	0.23063
1.8	0.013228	0.013491	0.77876	0.77876	0.20801	0.20775
1.9	0.011988	0.012211	0.79977	0.79978	0.18824	0.18801
2.0	0.010908	0.011100	0.81801	0.81801	0.17108	0.17089

Table 3: Comparison of ν_μ oscillation probabilities in vacuum and matter for the inverted mass hierarchy, Earth crust density = 2.6 g/cm^3, $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times$ 10^{-5} eV^2/c^4, $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4 and δ_{CP} = 0

Energy (GeV)	P($\nu_\mu \rightarrow \nu_e$)		P($\nu_\mu \rightarrow \nu_\mu$)		P($\nu_\mu \rightarrow \nu_\tau$)	
	Vacuum	Matter	Vacuum	Matter	Vacuum	Matter
0.3	0.016651	0.012718	0.91243	0.91427	0.070915	0.073016
0.4	0.047961	0.041880	0.28695	0.28747	0.66509	0.67065
0.5	0.052910	0.047892	0.019598	0.019587	0.92749	0.93252
0.6	0.047803	0.044104	0.021900	0.021756	0.93030	0.93414
0.7	0.040787	0.038101	0.11802	0.11787	0.84120	0.84403
0.8	0.034297	0.032325	0.23236	0.23222	0.73335	0.73545
0.9	0.028857	0.027381	0.33905	0.33895	0.63210	0.63367
1.0	0.024433	0.023306	0.43123	0.43115	0.54434	0.54555
1.1	0.020858	0.019980	0.50865	0.50859	0.47049	0.47143
1.2	0.017960	0.017265	0.57310	0.57305	0.40894	0.40969
1.3	0.015595	0.015037	0.62672	0.62669	0.35768	0.35828
1.4	0.013649	0.013194	0.67151	0.67148	0.31485	0.31533
1.5	0.012034	0.011657	0.70911	0.70909	0.27886	0.27926
1.6	0.010681	0.010366	0.74089	0.74087	0.24843	0.24876
1.7	0.0095379	0.0092726	0.76792	0.76791	0.22254	0.22282
1.8	0.0085654	0.0083395	0.79107	0.79105	0.20037	0.20061
1.9	0.0077316	0.0075376	0.81101	0.81010	0.18126	0.18146
2.0	0.0070119	0.0068441	0.82829	0.82829	0.16470	0.16487

2.5. Validation of the 3-flavour oscillation probabilities

The 3-flavour oscillation probabilities in vacuum were checked using an alternative formulation in equations 13.13 and 13.14 in the PDG review [11]:

$$P(\nu_\alpha \to \nu_\beta) = \sum_j |U_{\beta j}|^2 |U_{\alpha j}|^2 + 2 \sum_{j>k} |U_{\beta j} U^*_{\alpha j} U_{\alpha k} U^*_{\beta k}| \cos\left(\frac{\Delta m^2_{jk} L}{2E} - \phi_{\beta\alpha;jk} \right) \quad (8)$$

where $\phi_{\beta\alpha;jk}$ = arg($U_{\beta j} U^*_{\alpha j} U_{\alpha k} U^*_{\beta k}$). $P(\bar{\nu}_\alpha \to \bar{\nu}_\beta)$ is calculated by changing the sign of $\phi_{\beta\alpha;jk}$ in the argument of the cosine. The probabilities calculated from equation 8 agreed with those calculated from equation 1 to 14 significant figures.

Several checks of the oscillation probabilities in constant density matter are written into the code that calculates them:

1. The eigenvalues of $UMU^\dagger$ are checked by comparing their sum with the trace of the matrix.
2. Each normalised eigenvector is multiplied in turn by each of the 3 matrices formed by subtracting the eigenvalues from the real parts of the diagonal elements of $UMU^\dagger$, and a check is made of the differences between these products and a zero vector.
3. A check is made that the normalised eigenvectors of $UMU^\dagger$ are orthogonal by calculating the scalar product between them (the scalar product of two complex vectors is the product of the first vector with the complex conjugate of the second vector).
4. A check is also made that the Hermitian conjugate of the mixing matrix in matter is equal to the inverse of the matrix. This is done by multiplying the mixing matrix in matter by its Hermitian conjugate, and checking the differences between the product and an identity matrix.

The values of the oscillation probabilities in constant density matter were checked by comparing them with equivalent probabilities calculated by an independently-written Fortran program [13]. This Fortran program uses different algorithms, and calculates numerically the eigenvalues and eigenvectors of $UMU^\dagger$. Nevertheless there was very good agreement between the two calculations of the probabilities, with the fractional differences being $\approx 2 \times 10^{-6}$.

The values of the matter oscillation probabilities were also compared with those given by Prob3++ [22], and the fractional differences were $\approx$ 1-5 $\times 10^{-4}$. The comparison plots can be seen in [23].

2.6. Accuracy of the 3-flavour oscillation probabilities

As stated in section 2.5, the 3-flavour oscillation probabilities in vacuum agreed to 14 significant figures between the two formulations.

An estimate of the accuracy of the 3-flavour oscillation probabilities in constant density matter was made in three separate ways:

1. The matter probabilities for zero density were compared with the vacuum probabilities; there was good agreement, with the fractional differences being $\approx$ 2-5 $\times 10^{-6}$.
2. The probabilites given when calculating the eigenvectors with simultaneous equations 1 and 2 in $UMU^\dagger - \lambda I = 0$ were compared with the probabilities given when calculating the eigenvectors with simultaneous equations 2 and 3. The fractional differences between these two calculations were again $\approx$ 2-5 $\times 10^{-6}$.
3. The sum of three matter probabilities, for example P($\nu_\mu \to \nu_e$) + P($\nu_\mu \to \nu_\mu$) + P($\nu_\mu \to \nu_\tau$), was compared with 1.0; again there was good agreement, and the differences were $\approx 2 \times 10^{-6}$.

This means that the matter oscillation probabilities should be considered to be accurate to 5 decimal places. The accuracy of the matter probabilities is less than that of the vacuum probabilities due to the method of calculation of the eigenvectors. This involves calculations of the form

$$\frac{(ab - cd)}{(ef - gh)}$$

where a, b, c, etc. are elements of $UMU^\dagger - \lambda I$. Frequently ab is close in value to cd, which means that (ab - cd) is much smaller than ab or cd, and the same applies to (ef - gh). However the advantage of the method of calculation of the eigenvectors described in section 2.2 is that it allows the worst of these cancellation errors to be avoided; this can be done by calculating the eigenvectors using different pairs of simultaneous equations in $UMU^\dagger - \lambda I = 0$ for different combinations of the oscillation parameters. In practice, the eigenvectors are nearly always calculated using equations 1 and 2 in $UMU^\dagger - \lambda I = 0$, but equations 2 and 3 are used when $3.10 < \delta_{CP} < 3.18$ and $\sin^2(2\theta_{13}) > 10^{-8}$ in order to avoid such cancellation errors.

3. Updated predictions of single μ-like ring spectrum in Super-K

For 2-flavour oscillations, the predicted reconstructed single μ-like ring spectrum in Super-K depends only on $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$. However for the 3-flavour hypothesis, it also depends on $\sin^2(2\theta_{13})$, $\sin^2(2\theta_{12})$, Δm^2_{21}, δ_{CP} and the mass hierarchy. One other change from the 2-flavour analysis is that oscillated ν_e templates were added for the interaction modes CCQE, CC1π and CC other. An oscillated ν_e template for NC was not included since the ν_μ NC template effectively includes ν_e and ν_τ NC.

The predicted numbers of events in Super-K for each interaction mode are given in table 4 for GENIE and in table 5 for NEUT. The 2-flavour predictions are made using $\sin^2(2\theta_{23})$ = 1.0 and $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4. The nominal 3-flavour predictions are made with $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times$ 10^{-5} eV^2/c^4, δ_{CP} = 0 and Earth crust density = 2.6 g/cm^3; the second mass-squared difference is 2.32 $\times$ 10^{-3} eV^2/c^4 and this is input either as $|\Delta m^2_{32}|$ or as $|\Delta m^2_{FL}| = |m^2_3 - [(m^2_2 + m^2_1)/2]|$. The ranges of 3-flavour predictions in tables 4 and 5 represent the maximum and minimum numbers from 1000 random variations of the non-23 parameters within their uncertainties using the same values of $\sin^2(2\theta_{23})$ and the atmospheric mass-squared difference. When the the atmospheric Δm^2 is input as $|\Delta m^2_{32}|$, there are more ν_μ CCQE and ν_μ CC1π events in the 3-flavour inverted-hierarchy (IH) prediction and less in the 3-flavour normal-hierarchy (NH) prediction than in the 2-flavour prediction. This is because the ν_μ-survival probability is higher for the IH than the NH above the oscillation maximum and lower below the oscillation maximum (see figure 1). Since there are more events above the oscillation maximum than below it, the total number is larger for the IH than for the NH. This is true for any combination of the non-23 oscillation parameters. When the the atmospheric Δm^2 is input as $|\Delta m^2_{FL}|$, the 3-flavour predicted numbers of events are very similar to the 2-flavour numbers; this is because $|\Delta m^2_{31}|$ is equal to $|\Delta m^2_{32}|$ in the 2-flavour approximation, and $|\Delta m^2_{FL}|$ represents the average of these two mass-squared differences.

The predicted reconstructed single-ring μ-like Super-K spectrum using GENIE is shown in the upper plot of figure 4. In this plot, 3-flavour spectra are shown both for the value 2.32 $\times$ 10^{-3} eV^2/c^4 input as $|\Delta m^2_{32}|$ and for the same value input as $|\Delta m^2_{FL}|$ for both the NH and IH. The 2-flavour predicted spectrum is also shown for comparison. The lower plot shows the ratios of all four 3-flavour spectra to the 2-flavour spectrum. The equivalent plots using NEUT are shown in figure 5. These figures also show that, when the atmospheric Δm^2 is input as $|\Delta m^2_{32}|$, more events are predicted above the oscillation maximum and less below it for the IH than for the NH. When the the atmospheric Δm^2 is input as $|\Delta m^2_{FL}|$, they show that the 3-flavour predicted spectra are very similar to the 2-flavour spectra.

The effects of uncertainties in the non-23 oscillation parameters on the predicted reconstructed single-ring μ-like Super-K spectrum are shown in figures 6, 7, 8 and 9. For greater clarity, these effects are shown as ratios of the spectrum for $\pm1\sigma$ variations of a single oscillation parameter to that for the nominal value of that parameter, with the other five oscillation parameters being fixed. The uncertainties in $\sin^2(2\theta_{12})$ and Δm^2_{21} are taken from KamLAND combined with solar experiments [19], while those in $\sin^2(2\theta_{13})$ are computed as a weighted average of measurements from Daya Bay [20] and Reno [21].

By far the largest effect on the Super-K spectrum is from uncertainties in δ_{CP}: changing this parameter from 0 to π results in an increase of up to 5% in the spectrum above the oscillation maximum and a decrease of up to 3% below it for the NH. These changes in the spectrum are due to the changes in P($\nu_\mu \rightarrow \nu_\mu$) for different values of δ_{CP} shown in figure 3 and described in section 2. The effect of δ_{CP} on the predicted Super-K spectrum depends on the value of θ_{13} since the $e^{i\delta}$ terms in the PMNS matrix (equation 2) are always multiplied by $\sin(\theta_{13})$; in figure 7, $\sin^2(2\theta_{13})$ has the value 0.1. The effects of uncertainties in the other three non-23 parameters are smaller, and in all cases change the predicted Super-K spectrum by $\leq$ 0.5%. For the NH, an increase in the values of both $\sin^2(2\theta_{12})$ and Δm^2_{21} results in an increase in the predicted numbers of events at energies below the oscillation maximum, and a decrease in them above the oscillation maximum; these changes are due to the changes in P($\nu_\mu \rightarrow \nu_\mu$) for different values of $\sin^2(2\theta_{12})$ and Δm^2_{21} shown in table 1. The effects of δ_{CP}, $\sin^2(2\theta_{12})$ and Δm^2_{21} are all reversed for the inverted hierarchy. The effect of an increase in $\sin^2(2\theta_{13})$ is to increase the predicted number of events in Super-K for both hierarchies, while a decrease in this parameter results in a decrease in the predicted number. This is because an increase in $\sin^2(2\theta_{13})$ increases P($\nu_\mu \rightarrow \nu_\mu$) for a given value of $\sin^2(2\theta_{23})$ as shown in figure 2, i.e. more ν_μ survive for higher values of $\sin^2(2\theta_{13})$.

Table 4: Predicted numbers of single μ-like ring events in Super-K for each interaction mode using GENIE. The 2-flavour predictions are made using $\sin^2(2\theta_{23})$ = 1.0 and $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4. The nominal 3-flavour predictions are made with $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times$ 10^{-5} eV^2/c^4, δ_{CP} = 0 and Earth crust density = 2.6 g/cm^3; the second mass-squared difference is 2.32 $\times$ 10^{-3} eV^2/c^4 and this is input either as $|\Delta m^2_{32}|$ or as $|\Delta m^2_{FL}| = |m^2_3 - [(m^2_2 + m^2_1)/2]|$. The ranges of 3-flavour predictions represent the maximum and minimum numbers from 1000 random variations of the non-23 parameters within their uncertainties using the same values of $\sin^2(2\theta_{23})$ and the atmospheric mass-squared difference. NH = normal mass hierarchy, IH = inverted mass hierarchy.

Interaction mode	2-flavour	3-flavour $\|\Delta m^2_{32}\|$ (matter, NH)		3-flavour $\|\Delta m^2_{32}\|$ (matter, IH)		3-flavour $\|\Delta m^2_{FL}\|$ (matter, NH)		3-flavour $\|\Delta m^2_{FL}\|$ (matter, IH)	
		Nominal	Range	Nominal	Range	Nominal	Range	Nominal	Range
Total	28.39	28.14	28.06 - 28.43	28.87	28.56 - 28.98	28.51	28.44 - 28.83	28.45	28.17 - 28.52
ν_μ CCQE	16.52 (58.1%)	16.32 (58.0%)	16.27 - 16.49	16.79 (58.2%)	16.59 - 16.87	16.55 (58.0%)	16.50 - 16.76	16.51 (58.0%)	16.33 - 16.55
ν_μ CC1π	6.68 (23.6%)	6.56 (23.3%)	6.55 - 6.64	6.78 (23.5%)	6.70 - 6.81	6.68 (23.4%)	6.66 6.76	6.66 (23.4%)	6.58 - 6.68
ν_μ CCother	1.11 (3.9%)	1.10 (3.4%)	1.10 - 1.10	1.11 (3.8%)	1.11 - 1.11	1.11 (3.9%)	1.11 - 1.11	1.11 (3.9%)	1.10 - 1.11
ν_μ (ν_τ) NC	2.12 (7.5%)	2.12 (7.5%)	2.12 - 2.12	2.12 (7.3%)	2.12 -2.12	2.12 (7.4%)	2.12 - 2.12	2.12 (7.4%)	2.12 - 2.12
$\bar{\nu}_\mu$ CCQE	1.05 (3.7%)	1.05 (3.7%)	1.05 - 1.05	1.07 (3.7%)	1.07 -1.08	1.07 (3.7%)	1.06 - 1.07	1.06 (3.7%)	1.05 - 1.06
$\bar{\nu}_\mu$ CC1π	0.63 (2.3%)	0.64 (2.3%)	0.64 - 0.64	0.65 (2.3%)	0.65 - 0.65	0.64 (2.3%)	0.64 - 0.65	0.64 (2.3%)	0.64 - 0.64
$\bar{\nu}_\mu$ CCother	0.11 (0.4%)	0.11 (0.4%)	0.11 - 0.11	0.11 (0.4%)	0.11 - 0.11	0.11 (0.4%)	0.11 - 0.11	0.11 (0.4%)	0.11 - 0.11
$\bar{\nu}_\mu$ ($\bar{\nu}_\tau$) NC	0.12 (0.4%)	0.12 (0.4%)	0.12 - 0.12	0.12 (0.4%)	0.12 - 0.12	0.12 (0.4%)	0.12 - 0.12	0.12 (0.4%)	0.12 - 0.12
ν_e CCQE (background)	0.007 (0.02%)	0.006 (0.02%)	0.006 - 0.007	0.007 (0.02%)	0.006 - 0.007	0.006 (0.02%)	0.006 - 0.007	0.007 (0.02%)	0.006 - 0.007
ν_e CC1π (background)	0.004 (0.01%)	0.004 (0.01%)	0.004 - 0.004	0.004 (0.01%)	0.004 - 0.004	0.004 (0.01%)	0.004 - 0.004	0.004 (0.01%)	0.004 - 0.004
ν_e CCother (background)	0.001 (0.003%)	0.001 (0.003%)	0.001 - 0.001	0.001 (0.003%)	0.001 - 0.001	0.001 (0.003%)	0.001 - 0.001	0.001 (0.003%)	0.001 - 0.001
ν_e NC (background)	0.06 (0.2%)	0.06 (0.2%)	0.06 - 0.06	0.06 (0.2%)	0.06 - 0.06	0.06 (0.2%)	0.06 - 0.06	0.06 (0.2%)	0.06 - 0.06
ν_e CCQE (oscillated)		0.04 (0.1%)	0.02 - 0.07	0.04 (0.1%)	0.02 - 0.06	0.04 (0.1%)	0.02 - 0.07	0.04 (0.1%)	0.01 - 0.06
ν_e CC1π (oscillated)		0.01 (0.04%)	0.004 - 0.02	0.01 (0.03%)	0.004 - 0.017	0.01 (0.05%)	0.005 - 0.02	0.01 (0.04%)	0.005 - 0.017
ν_e CCother (oscillated)		0.0006 (0.002%)	0.0002 - 0.0008	0.0004 (0.001%)	0.0002 - 0.0007	0.0006 (0.002%)	0.0002 - 0.0009	0.0004 (0.001%)	0.0002 - 0.0007

Table 5: Predicted numbers of single μ-like ring events in Super-K for each interaction mode using NEUT. The 2-flavour predictions are made using $\sin^2(2\theta_{23})$ = 1.0 and $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4. The nominal 3-flavour predictions are made with $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times$ 10^{-5} eV^2/c^4, δ_{CP} = 0 and Earth crust density = 2.6 g/cm^3; the second mass-squared difference is 2.32 $\times$ 10^{-3} eV^2/c^4 and this is input either as $|\Delta m^2_{32}|$ or as $|\Delta m^2_{FL}| = |m^2_3 - [(m^2_2 + m^2_1)/2]|$. The ranges of 3-flavour predictions represent the maximum and minimum numbers from 1000 random variations of the non-23 parameters within their uncertainties using the same values of $\sin^2(2\theta_{23})$ and the atmospheric mass-squared difference. NH = normal mass hierarchy, IH = inverted mass hierarchy.

Interaction mode	2-flavour	3-flavour $\|\Delta m^2_{32}\|$ (matter, NH)		3-flavour $\|\Delta m^2_{32}\|$ (matter, IH)		3-flavour $\|\Delta m^2_{FL}\|$ (matter, NH)		3-flavour $\|\Delta m^2_{FL}\|$ (matter, IH)	
		Nominal	Range	Nominal	Range	Nominal	Range	Nominal	Range
Total	29.18	28.86	28.79 - 29.26	29.74	29.40 - 29.86	29.31	29.23 - 29.70	29.23	28.92 - 29.33
ν_μ CCQE	16.73 (57.3%)	16.49 (57.1%)	16.45 - 16.75	17.08 (57.5%)	16.85 - 17.17	16.79 (57.3%)	16.73 - 17.05	16.74 (57.3%)	16.54 - 16.80
ν_μ CC1π	7.01 (24.0%)	6.90 (23.9%)	6.87 - 6.99	7.14 (24.0%)	7.05 - 7.16	7.02 (24.0%)	7.00 - 7.11	7.01 (24.0%)	6.93 - 7.03
ν_μ CCother	1.80 (6.2%)	1.79 (6.2%)	1.79 - 1.80	1.81 (6.1%)	1.80 - 1.81	1.80 (6.1%)	1.80 - 1.81	1.80 (6.2%)	1.79 - 1.80
ν_μ (ν_τ) NC	1.68 (5.7%)	1.69 (5.8%)	1.69 - 1.69	1.69 (5.7%)	1.69 - 1.69	1.69 (5.8%)	1.69 - 1.69	1.69 (5.8%)	1.69 - 1.69
$\bar{\nu}_\mu$ CCQE	1.11 (3.8%)	1.10 (3.8%)	1.10 - 1.11	1.13 (3.8%)	1.12 - 1.13	1.11 (3.8%)	1.11 - 1.12	1.11 (3.8%)	1.10 - 1.11
$\bar{\nu}_\mu$ CC1π	0.51 (1.7%)	0.51 (1.8%)	0.50 - 0.51	0.51 (1.7%)	0.51 - 0.52	0.51 (1.7%)	0.51 - 0.51	0.51 (1.7%)	0.51 0.51
$\bar{\nu}_\mu$ CCother	0.17 (0.6%)	0.17 (0.6%)	0.17 - 0.17	0.17 (0.6%)	0.17 - 0.17	0.17 (0.6%)	0.17 - 0.17	0.17 (0.6%)	0.17 - 0.17
$\bar{\nu}_\mu$ ($\bar{\nu}_\tau$) NC	0.093 (0.3%)	0.093 (0.3%)	0.093 - 0.093	0.093 (0.3%)	0.093 - 0.093	0.093 (0.3%)	0.093 - 0.093	0.093 (0.3%)	0.093 0.093
ν_e CCQE (background)	0.008 (0.03%)	0.007 (0.02%)	0.007 - 0.007	0.007 (0.02%)	0.007 - 0.007	0.007 (0.02%)	0.007 - 0.007	0.007 (0.02%)	0.007 - 0.007
ν_e CC1π (background)	0.005 (0.02%)	0.004 (0.02%)	0.004 - 0.004	0.004 (0.01%)	0.004 - 0.004	0.004 (0.01%)	0.004 0.004	0.004 (0.01%)	0.004 - 0.004
ν_e CCother (background)	0.002 (0.006%)	0.001 (0.005%)	0.001 - 0.001	0.001 (0.005%)	0.001 - 0.001	0.001 (0.005%)	0.001 - 0.001	0.001 (0.005%)	0.001 - 0.001
ν_e NC (background)	0.05 (0.2%)	0.05 (0.2%)	0.05 - 0.05	0.05 (0.2%)	0.05 - 0.05	0.05 (0.2%)	0.05 - 0.05	0.05 (0.2%)	0.05 - 0.05
ν_e CCQE (oscillated)		0.04 (0.1%)	0.02 - 0.07	0.04 (0.1%)	0.02 - 0.06	0.04 (0.1%)	0.02 - 0.07	0.04 (0.1%)	0.02 - 0.06
ν_e CC1π (oscillated)		0.015 (0.04%)	0.006 - 0.023	0.01 (0.04%)	0.006 - 0.020	0.01 (0.05%)	0.005 - 0.022	0.01 (0.04%)	0.005 - 0.021
ν_e CCother (oscillated)		0.001 (0.005%)	0.0007 - 0.0022	0.001 (0.004%)	0.0006 - 0.0019	0.001 (0.005%)	0.0005 - 0.0021	0.001 (0.004%)	0.0005 - 0.0020

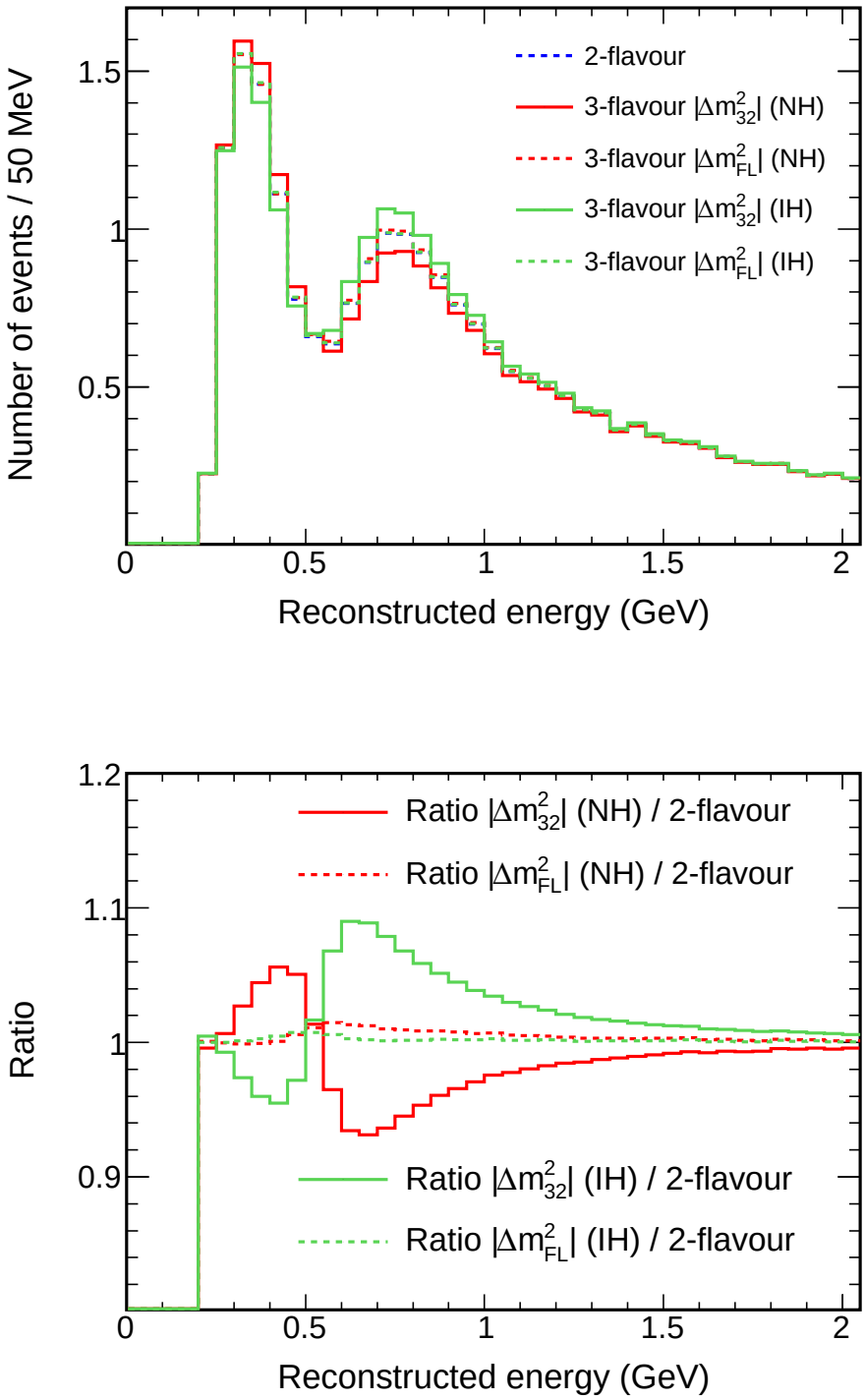


Figure 4: Predicted reconstructed single μ-like ring spectra in Super-K for the 2-flavour and 3-flavour (in matter) hypotheses using GENIE (top); for the 3-flavour spectra, the value 2.32 $\times$ 10^{-3} eV^2/c^4 input either as $|\Delta m^2_{32}|$ or as $|\Delta m^2_{FL}| = |m^2_3 - [(m^2_2 + m^2_1)/2]|$. The ratios of the 3-flavour spectra to the 2-flavour spectrum are shown in the bottom plot; $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, δ_{CP} = 0, Earth crust density = 2.6 g/cm^3, NH = normal mass hierarchy and IH = inverted mass hierarchy.

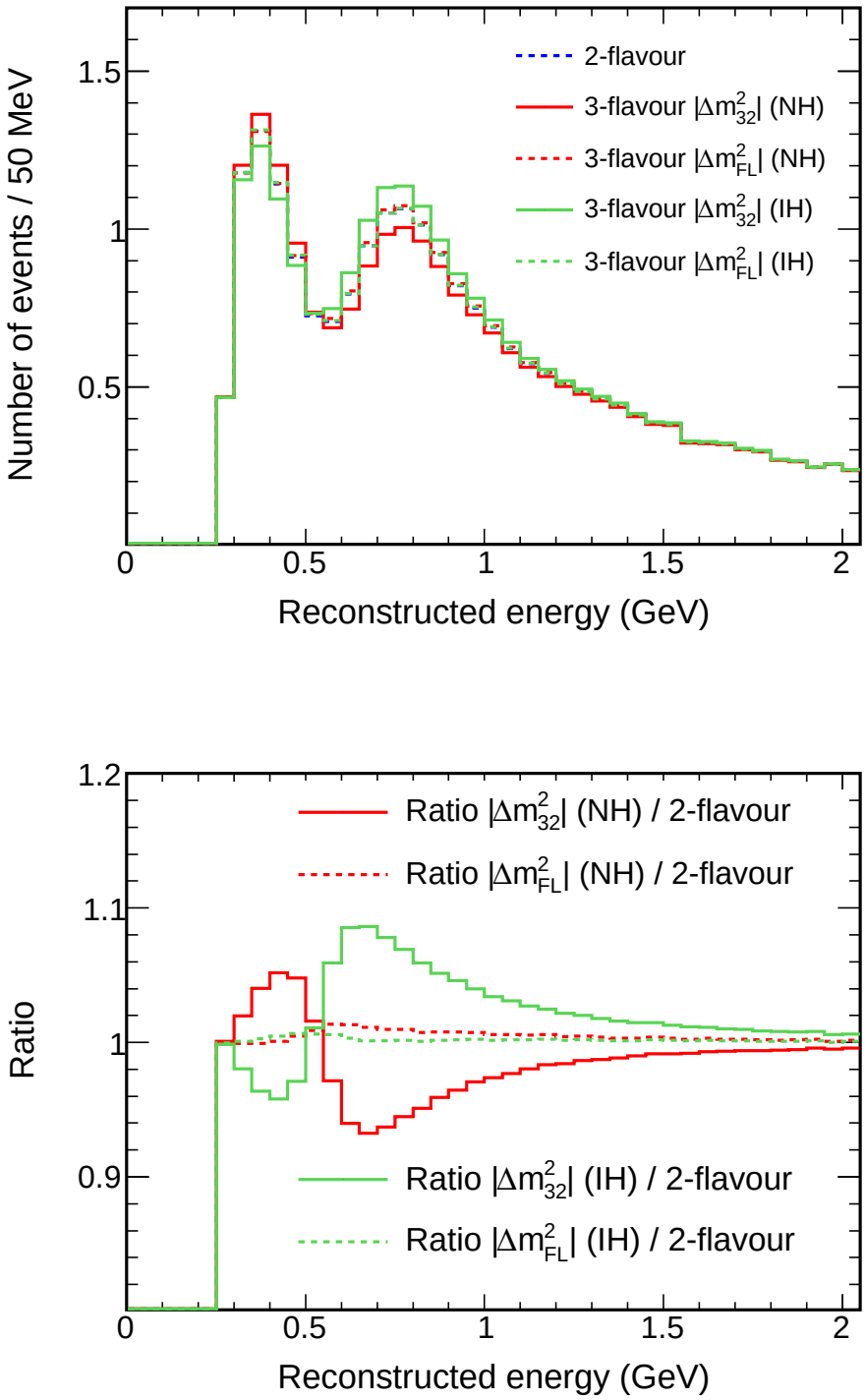


Figure 5: Predicted reconstructed single μ-like ring spectra in Super-K for the 2-flavour and 3-flavour (in matter) hypotheses using NEUT (top); for the 3-flavour spectra, the value 2.32 $\times$ 10^{-3} eV^2/c^4 input either as $|\Delta m^2_{32}|$ or as $|\Delta m^2_{FL}| = |m^2_3 - [(m^2_2 + m^2_1)/2]|$. The ratios of the 3-flavour spectra to the 2-flavour spectrum are shown in the bottom plot; $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, δ_{CP} = 0, Earth crust density = 2.6 g/cm^3, NH = normal mass hierarchy and IH = inverted mass hierarchy.

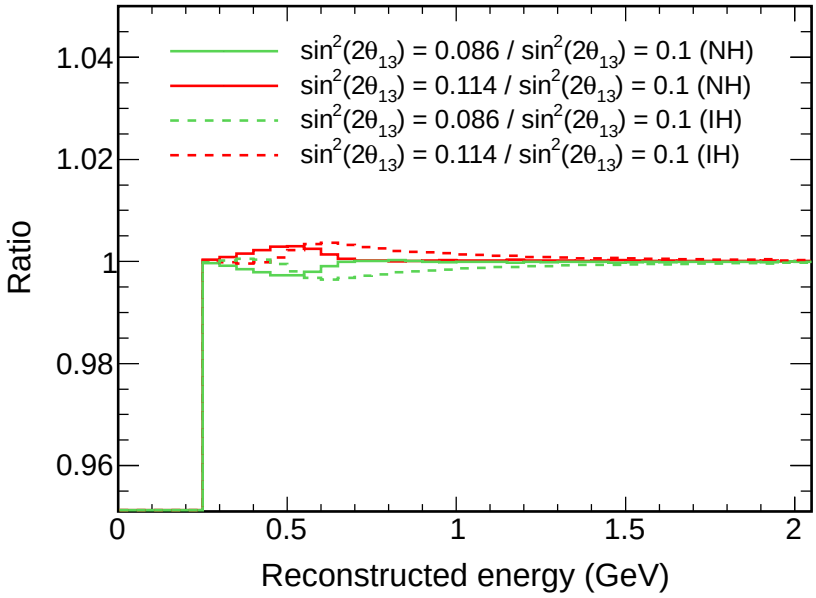


Figure 6: Ratios of predicted reconstructed single μ-like ring spectra in Super-K for $\pm 1\sigma$ variations in $\sin^2(2\theta_{13})$ to those for the nominal value using NEUT; $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4, δ_{CP} = 0, Earth crust density = 2.6 g/cm^3, NH = normal mass hierarchy and IH = inverted mass hierarchy

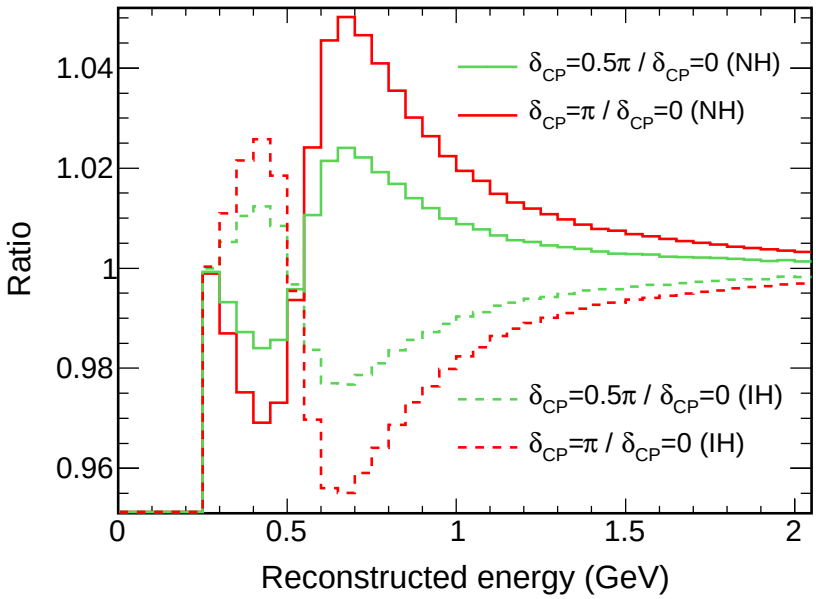


Figure 7: Ratios of predicted reconstructed single μ-like ring spectra in Super-K for $\delta_{CP} = \pi/2$ and $\delta_{CP} = \pi$ to those for δ_{CP} = 0 using NEUT; $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4,$|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4, Earth crust density = 2.6 g/cm^3, NH = normal mass hierarchy and IH = inverted mass hierarchy

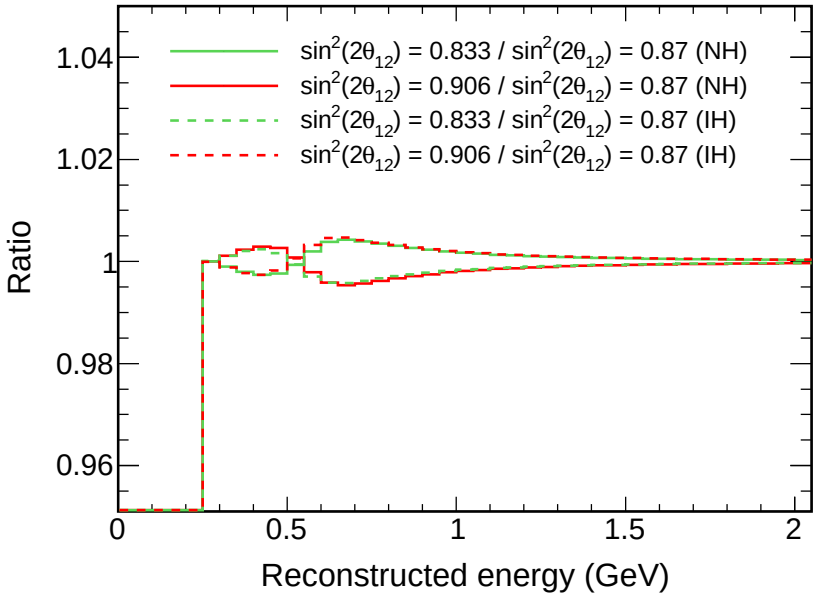


Figure 8: Ratios of predicted reconstructed single μ-like ring spectra in Super-K for $\pm 1\sigma$ variations in $\sin^2(2\theta_{12})$ to those for the nominal value using NEUT; $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4, δ_{CP} = 0, Earth crust density = 2.6 g/cm^3, NH = normal mass hierarchy and IH = inverted mass hierarchy

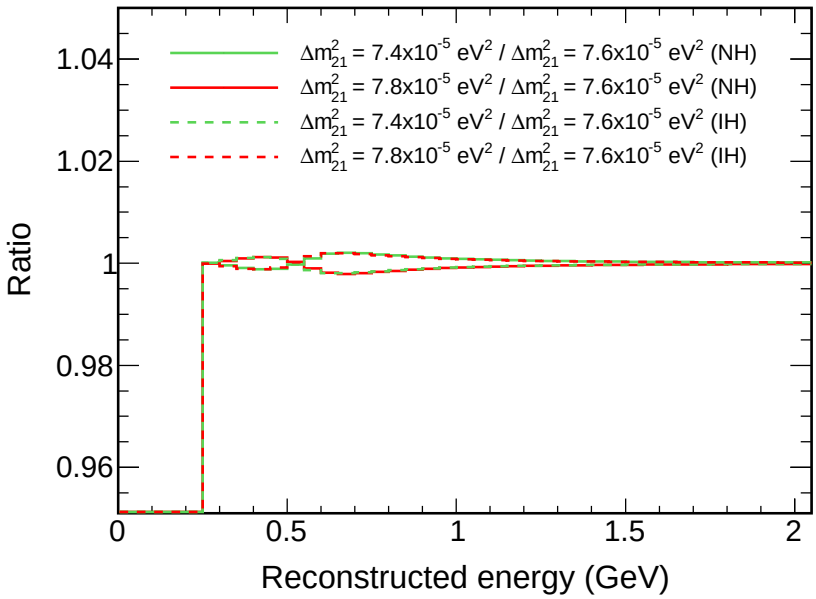


Figure 9: Ratios of predicted reconstructed single μ-like ring spectra in Super-K for $\pm 1\sigma$ variations in Δm^2_{21} to those for the nominal value using NEUT; $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, $\sin^2(2\theta_{23})$ = 1.0, $|\Delta m^2_{32}|$ = 2.32 $\times$ 10^{-3} eV^2/c^4, δ_{CP} = 0, Earth crust density = 2.6 g/cm^3, NH = normal mass hierarchy and IH = inverted mass hierarchy

Table 6: Best-fit values of $\sin^2(2\theta_{23})$ and the atmospheric Δm^2 ($|\Delta m^2_{32}|$ or $|\Delta m^2_{FL}|$) for GENIE and NEUT and the 3-flavour and 2-flavour analyses. When making the fit, the non-23 oscillation parameters are fixed at $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, Δm^2_{21} = 7.6 $\times$ 10^{-5} eV^2/c^4 and δ_{CP} = 0. The Earth crust density is taken to be 2.6 g/cm^3, and NH = normal mass hierarchy, IH = inverted mass hierarchy.

Generator	Parameter	3-flavour $\|\Delta m^2_{32}\|$ (NH)	3-flavour $\|\Delta m^2_{32}\|$ (IH)	3-flavour $\|\Delta m^2_{FL}\|$ (NH)	3-flavour $\|\Delta m^2_{FL}\|$ (IH)	2-flavour
GENIE	$\sin^2(2\theta_{23})$	0.98	0.98	0.98	0.98	0.97
	$\|\Delta m^2_{32}\|$ (10^{-3} eV^2/c^4)	2.6	2.6	2.6	2.6	2.6
	χ^2/ndf (ndf = 198)	0.31	0.31	0.31	0.31	0.31
NEUT	$\sin^2(2\theta_{23})$	0.99	0.99	0.99	0.99	0.98
	$\|\Delta m^2_{32}\|$ (10^{-3} eV^2/c^4)	2.6	2.7	2.7	2.6	2.6
	χ^2/ndf (ndf = 198)	0.31	0.31	0.31	0.31	0.31

4. 3-flavour ν_μ-disappearance fits to the Run 1+2 dataset

The best-fit values of $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ are obtained using a binned likelihood-ratio method in which the $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ are varied in the input to the calculation of the expected single μ-like ring reconstructed energy spectrum until the quantity

$$2 \sum_{E_{RECO}} \left[N^{data}_{SK} \ln \left(\frac{N^{data}_{SK}}{N^{exp}_{SK}} \right) + (N^{exp}_{SK} - N^{data}_{SK}) \right] \quad (9)$$

is minimised. Further details of this method are given in sections 3.1 and 3.2 in [2].

In a 3-flavour ν_μ-disappearance fit, it is necessary to make a choice for the non-23 oscillation parameters. Possible options for these parameters include

1. All 6 parameters, i.e. $\sin^2(2\theta_{12})$, $\sin^2(2\theta_{13})$, $\sin^2(2\theta_{23})$, Δm^2_{21}, $|\Delta m^2_{32}|$ and δ_{CP} can float freely.
2. All 6 parameters can float freely, but deviations in values of $\sin^2(2\theta_{12})$, $\sin^2(2\theta_{13})$ and Δm^2_{21} from values measured by other experiments are penalised.
3. $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ can float freely, but $\sin^2(2\theta_{12})$, $\sin^2(2\theta_{13})$, Δm^2_{21} and δ_{CP} are fixed.

In the present analysis, option 3 is used, and the values of $\sin^2(2\theta_{12})$, $\sin^2(2\theta_{13})$, Δm^2_{21} and δ_{CP} are fixed at 0.87, 0.1, 7.6 $\times$ 10^{-5} eV^2/c^4 and 0.0 respectively when finding the best-fit values of $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$. In future analyses by the Rutherford-Oxford-Lancs group, the uncertainties in the non-23 parameters will be taken into account in the confidence contours; however this is not done in the present analysis since their effects on the contours were found to be negligible (see section 5).

The 3-flavour best-fit points for both mass hierarchies using both GENIE and NEUT are shown in table 6; the 2-flavour best-fit points [1] are also given for comparison. The best-fit value of $\sin^2(2\theta_{23})$ is higher for the 3-flavour than for the 2-flavour analysis due to the effect of $\sin^2(2\theta_{13})$: in the 3-flavour analysis, $\sin^2(2\theta_{13})$ is fixed at 0.1 in the fitter whereas in the 2-flavour analysis its value is approximated to be 0.0. As shown in figure 2, more ν_μ survive near the oscillation maximum when $\sin^2(2\theta_{13})$ = 0.1 than when $\sin^2(2\theta_{13})$ = 0 for a given value of $\sin^2(2\theta_{23})$. This means that, in the 3-flavour analysis, it is necessary to increase the value of $\sin^2(2\theta_{23})$ in order to fit the data. The best-fit value of $|\Delta m^2_{32}|$ (when it is used as the atmospheric mass-squared difference) is higher for the IH than for the NH when using NEUT. This is because the oscillation dip in the ν_μ-survival probability is at lower energy for the IH than for the NH at a given value of $|\Delta m^2_{32}|$ (see figure 1). In order to fit the oscillation dip at a certain energy in the real data, it is necessary to choose a higher value of $|\Delta m^2_{32}|$ for the IH than for the NH. This difference is not seen for the best-fit values of $|\Delta m^2_{32}|$ when using GENIE, though it is visible in the confidence contours; it is possible that the difference in best-fit values is absorbed by the relatively coarse binning used (the grid points are separated by 1×10^{-4} eV^2/c^4).

The reconstructed energy distribution of the 31 data events in the Run 1+2 dataset is shown in figures 10 and 11. For comparison, the spectra for no disappearance and best-fit 3-flavour oscillations are shown for GENIE in figure 10 and for NEUT in figure 11. These figures use a variable binning scheme which gives a clearer representation of these spectra; howvever the actual analysis was done in equal-sized bins of width 50 MeV. Best-fit spectra with oscillations are shown in these figures for both the normal and inverted hierarchies, but they cannot be separated by eye. In order to show the small differences between them, the ratio of the best-fit spectrum for the normal hierarchy to that for the inverted hierarchy is shown using NEUT in figure 12. In figures 10, 11 and 12, the atmospheric Δm^2 is input as $|\Delta m^2_{32}|$. The ratio of the best-fit oscillated Super-K spectrum for the atmospheric Δm^2 input as $|\Delta m^2_{32}|$ to the best-fit spectrum for the atmospheric Δm^2 input as $|\Delta m^2_{FL}| = |m^2_3 - [(m^2_2 + m^2_1)/2]|$ is shown using NEUT for both hierarchies in figure 13.

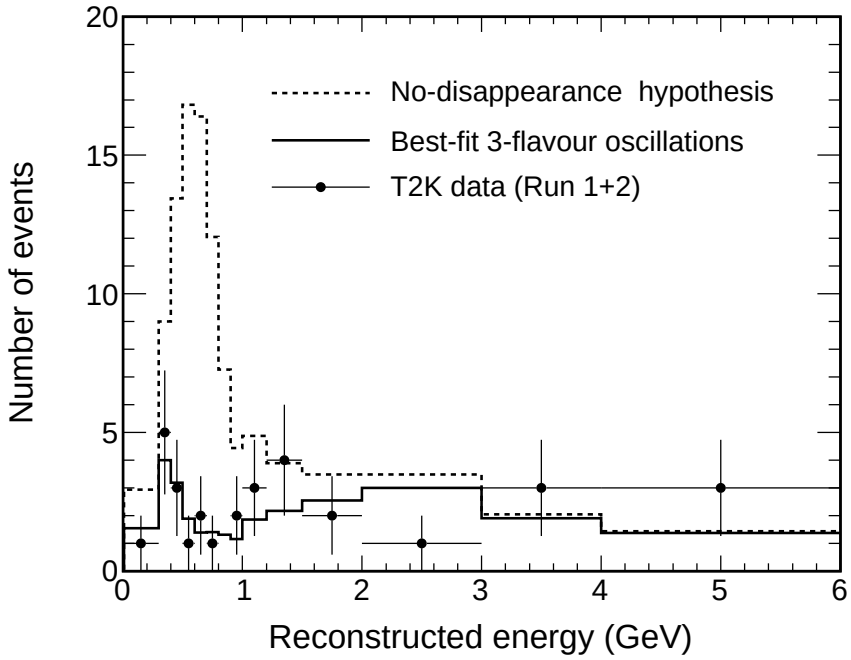


Figure 10: Reconstructed energy spectrum of the 31 Run 1+2 data events compared with the expected spectra in Super-K without disappearance and with best-fit 3-flavour oscillations (GENIE); the spectrum for best-fit oscillations uses $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, δ_{CP} = 0 and Earth crust density = 2.6 g/cm^3. The atmospheric Δm^2 is input as $|\Delta m^2_{32}|$. A variable binning scheme is used here for the purpose of illustration only; the actual analysis used equal-sized 50 MeV bins.

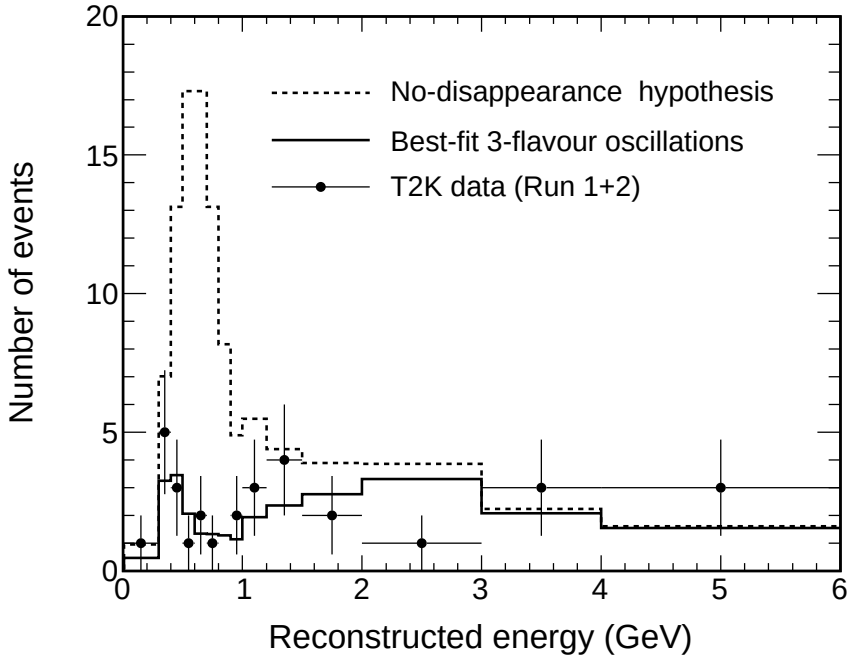


Figure 11: Reconstructed energy spectrum of the 31 Run 1+2 data events compared with the expected spectra in Super-K without disappearance and with best-fit 3-flavour oscillations (NEUT); the spectrum for best-fit oscillations uses $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, δ_{CP} = 0 and Earth crust density = 2.6 g/cm^3. The atmospheric Δm^2 is input as $|\Delta m^2_{32}|$. A variable binning scheme is used here for the purpose of illustration only; the actual analysis used equal-sized 50 MeV bins.

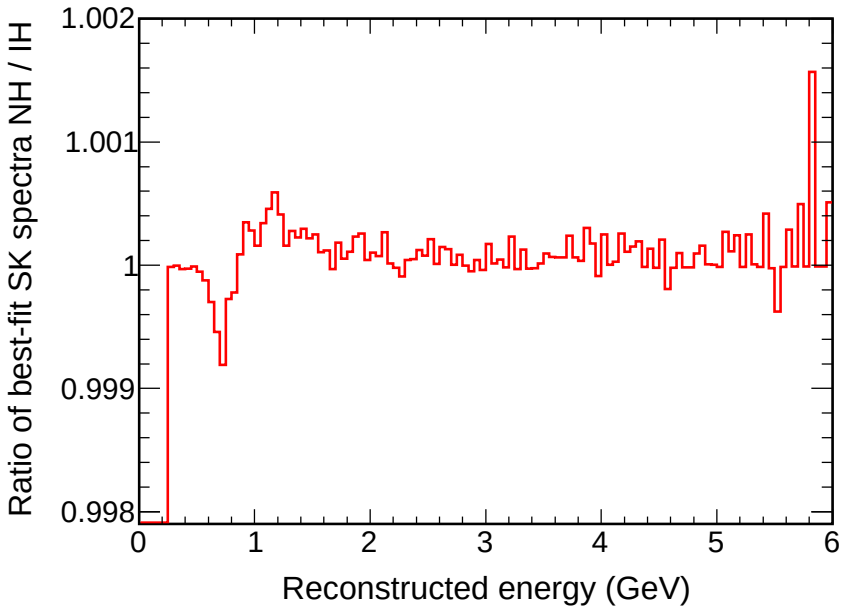


Figure 12: Ratio of best-fit 3-flavour reconstructed Super-K energy spectrum assuming the normal mass hierarchy to that assuming the inverted mass hierarchy using NEUT; $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, δ_{CP} = 0 and Earth crust density = 2.6 g/cm^3. The atmospheric Δm^2 is input as $|\Delta m^2_{32}|$.

The uncertainties in the best-fit values of $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ due to uncertainties in the values of the non-23 oscillation parameters were estimated by creating 50000 fake datasets for each mass hierarchy. These fake datasets did not include statistical fluctuations, and they were made using random input values of $\sin^2(2\theta_{12})$, $\sin^2(2\theta_{13})$ and Δm^2_{21} drawn from Gaussian distributions with means equal to the nominal values and standard deviations equal to the 1σ errors. The uncertainties in $\sin^2(2\theta_{12})$ and Δm^2_{21} are taken from KamLAND combined with solar experiments [19], while those in $\sin^2(2\theta_{13})$ are computed as a weighted average of measurements from Daya Bay [20] and Reno [21]. The input values of δ_{CP} were randomly drawn from a uniform distribution between -π and +π. The input value of $\sin^2(2\theta_{23})$ was fixed at 0.9 (to allow an estimate of both plus and minus uncertainties) and the input value of $|\Delta m^2_{32}|$ was fixed at 2.32 $\times$ 10^{-3} eV^2/c^4.

When fitting the fake datasets to obtain the best-fit points, the expected numbers of single μ-like ring events in Super-K with oscillations were again calculated with fixed values of $\sin^2(2\theta_{12})$, $\sin^2(2\theta_{13})$, Δm^2_{21} and δ_{CP}, while $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ were allowed to vary in order to find their best-fit values. When all the non-23 parameters were varied together, the standard deviations of the best-fit values of $\sin^2(2\theta_{23})$ were 0.5% (NH) and 0.4% (IH), and those of the best-fit values of $|\Delta m^2_{32}|$ were 0.7% (NH) and 0.4% (IH). Additional fake datasets were made in which only one non-23 parameter was varied at a time, with the other parameters being fixed at their default values. The results are given in table 7 which shows that the non-23 uncertainties in the best-fit points are almost entirely due to uncertainties in δ_{CP}. As a comparison, the uncertainties in the best-fit points due to the combined systematic uncertainties were considerably larger at 4.3% (NH) and 4.9% (IH) for $\sin^2(2\theta_{23})$ and 4.8% (NH) and 5.1% (IH) for $|\Delta m^2_{32}|$.

In order to show these uncertainties in the best-fit points graphically, they were also estimated by varying each non-23 parameter by $\pm 1\sigma$ in a single toy experiment without statistical fluctuations. Once more the input value of $\sin^2(2\theta_{23})$ was fixed at 0.9 and the input value of $|\Delta m^2_{32}|$ was fixed at 2.32 $\times$ 10^{-3} eV^2/c^4. The results are shown in figure 14, and the uncertainties in the best-fit points are again almost entirely due to uncertainties in δ_{CP}.

Since it is not curremtly possible to define a $\pm 1\sigma$ error in δ_{CP}, the effects of varying it on the best-fit values of $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ are plotted in figure 15. δ_{CP} can affect $|\Delta m^2_{32}|$ only (for example if it changes from 0.0 to π) or $\sin^2(2\theta_{23})$ only (for example if it changes from 0.0 to $\frac{3\pi}{4}$) or both (for example if it changes from 0.0 to $\frac{\pi}{2}$).

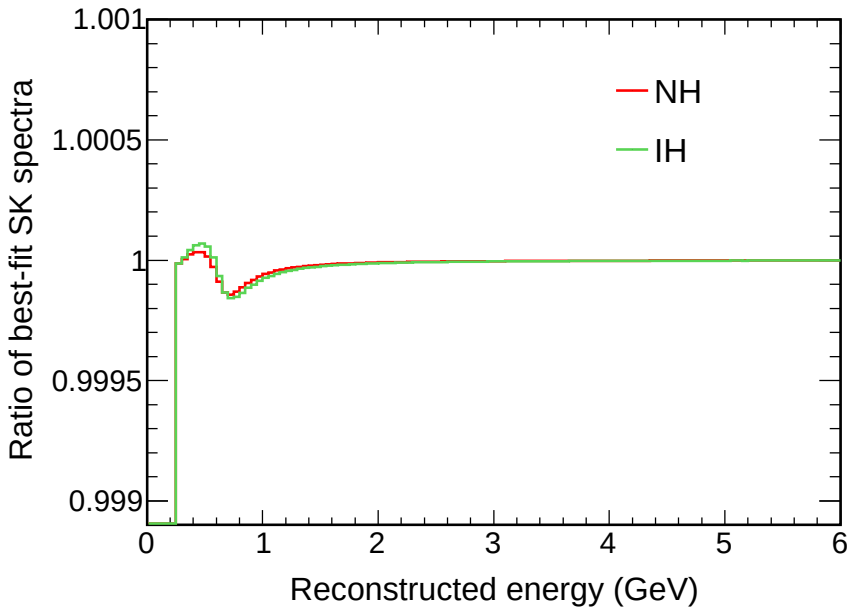
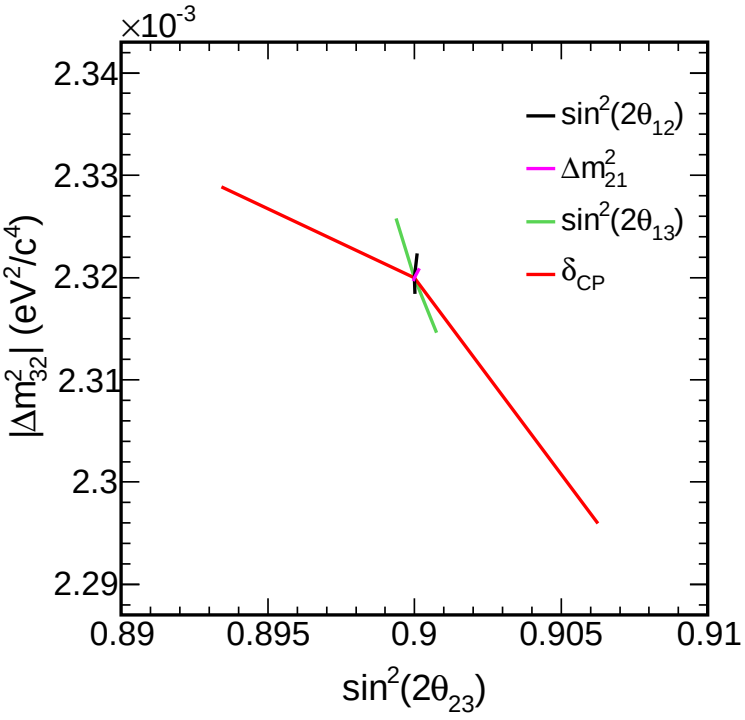


Figure 13: Ratio of best-fit 3-flavour reconstructed Super-K energy spectrum for the atmospheric Δm^2 input as $|\Delta m^2_{32}|$ to the best-fit spectrum for the atmospheric Δm^2 input as $|\Delta m^2_{FL}| = |m^2_3 - [(m^2_2 + m^2_1)/2]|$ using NEUT; $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, Δm^2_{21} = 7.6 $\times 10^{-5}$ eV^2/c^4, δ_{CP} = 0, Earth crust density = 2.6 g/cm^3, NH = normal mass hierarchy, IH = inverted mass hierarchy.

Table 7: Uncertainties in the best-fit values of $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ due to uncertainties in the non-23 oscillation parameters using NEUT. These are varied using random numbers drawn from Gaussian distributions with means equal to the nominal values and standard deviations equal to the 1σ errors, except for δ_{CP} which is varied using a random number drawn from a uniform distribution between -π and +π. When fixed these parameters have the values $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, Δm^2_{21} = 7.6 $\times$ 10^{-5} eV^2/c^4, δ_{CP} = 0. The Earth crust density is taken to be 2.6 g/cm^3, and NH = normal mass hierarchy, IH = inverted mass hierarchy.

Parameter	Uncertainty in $\sin^2(2\theta_{23})$ (NH)	Uncertainty in $\lvert\Delta m^2_{32}\rvert$ (NH)	Uncertainty in $\sin^2(2\theta_{23})$ (IH)	Uncertainty in $\lvert\Delta m^2_{32}\rvert$ (IH)
$\sin^2(2\theta_{13})$	0.08%	0.2%	0.01%	0.05%
δ_{CP}	0.5%	0.6%	0.35%	0.4%
$\sin^2(2\theta_{12})$	0.004%	0.09%	0.007%	0.09%
Δm^2_{21}	0.01%	0.04%	0.009%	0.05%
All non-23 parameters	0.5%	0.7%	0.4%	0.4%
All systematics	4.3%	4.8%	4.9%	5.1%



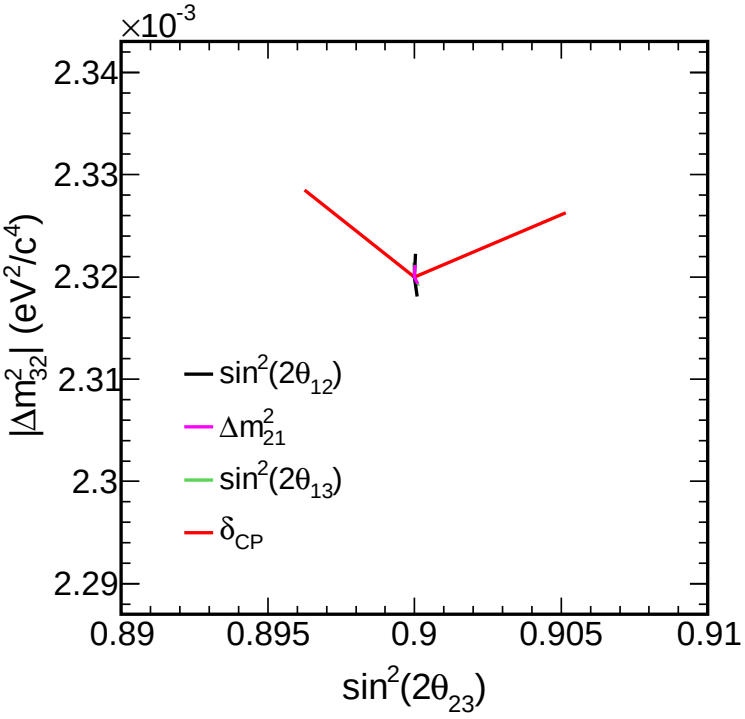
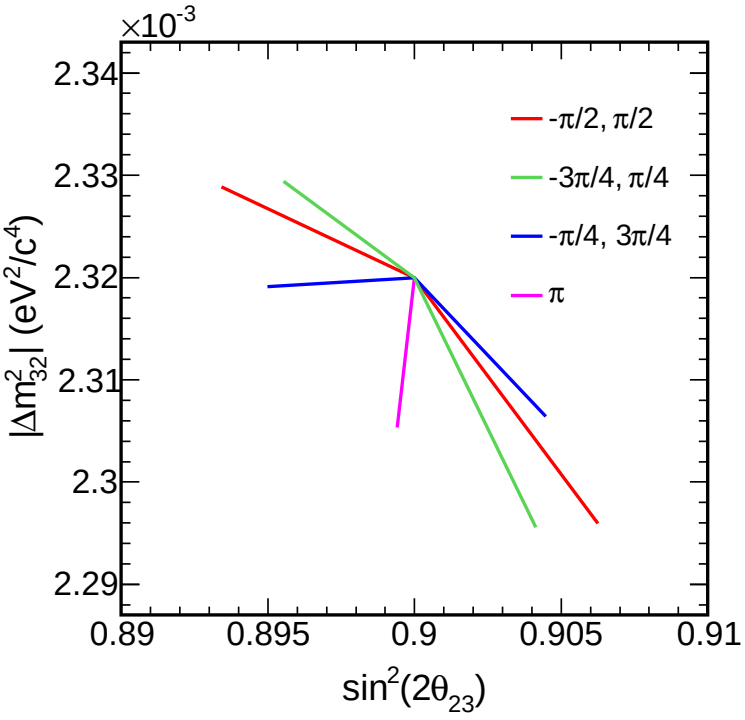


Figure 14: Uncertainties in the best-fit values of $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ due to uncertainties in the non-23 oscillation parameters for the normal mass hierarchy (top) and the inverted mass hierarchy (bottom) using NEUT; uncertainties in δ_{CP} are shown for changes from 0 to +π/2 and -π/2.



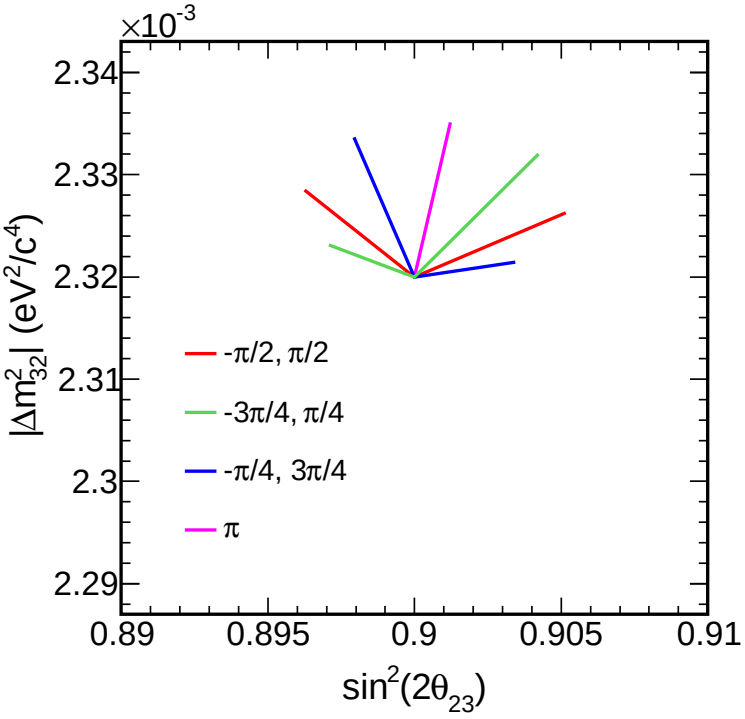


Figure 15: Uncertainties in the best-fit values of $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ due to uncertainties in δ_{CP} for the normal mass hierarchy (top) and the inverted mass hierarchy (bottom) using NEUT

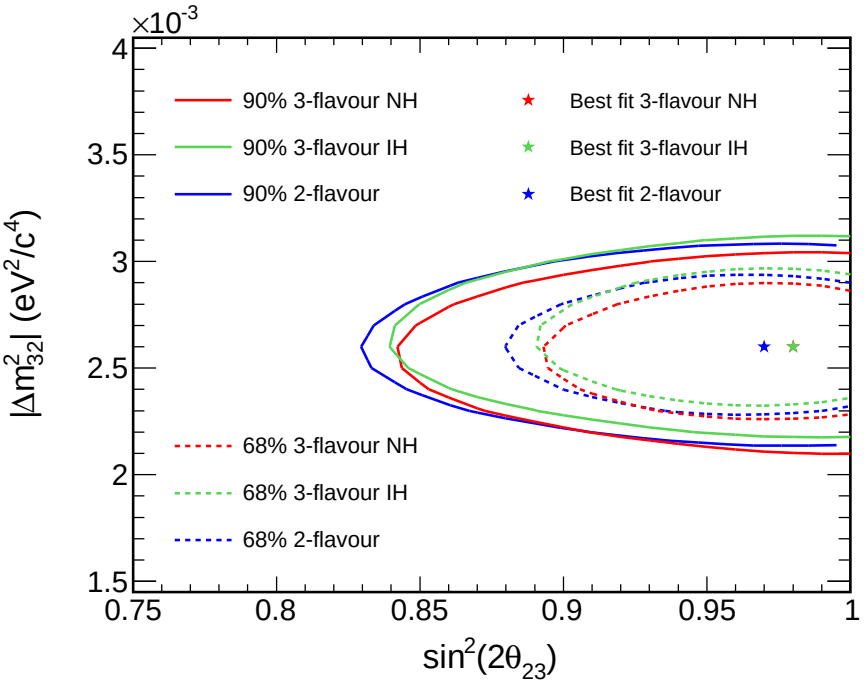


Figure 16: Comparison of confidence contours for $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ between 2-flavour, 3-flavour (NH) and 3-flavour (IH) using GENIE; the 3-flavour best-fit points are identical for the NH and IH. All contours include both statistical and systematic uncertainties. The non-23 oscillation parameters are fixed.

5. 3-flavour ν_μ-disappearance confidence contours for the Run 1+2 dataset

Confidence contours for the oscillation parameters $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ are drawn using the method of Feldman and Cousins [16]. Details of its implementation in the ν_μ-disappearance analysis are given in section 3.4 in [2].

A comparison of the stat + syst 68% and 90% confidence regions between the 2-flavour [1] and 3-flavour analyses is shown using GENIE in figure 16 and using NEUT in figure 17. In these figures the non-23 oscillation parameters are fixed. The 3-flavour contours are narrower in $\sin^2(2\theta_{23})$ than the 2-flavour contours, and this is because the 3-flavour best-fit points are nearer to $\sin^2(2\theta_{23})$ = 1.0 than the 2-flavour best-fit points. The 3-flavour contours for the IH include higher values of $|\Delta m^2_{32}|$ than those for the NH it is necessary to choose a higher value of $|\Delta m^2_{32}|$ for the IH than for the NH in order to fit the oscillation dip at a certain energy in the real data.

It was found that uncertainties in the non-23 oscillation parameters had a neglible effect on the 3-flavour confidence contours. This was checked by creating 10000 toy MC experiments using NEUT for each of a few oscillation parameter points on or near the 3-flavour 68% and 90% contours for the NH only. Examples of the cumulative distributions of $\Delta\chi^2$ are given for a point on the physical boundary in figure 18 and a point away from the physical boundary in figure 19. In both cases the distributions including the non-23 uncertainties are negligibly different from those without them. For this reason, it was decided not to remake the Feldman-Cousins contours including the non-23 uncertainties for the whole grid of oscillation parameter points for this analysis. However in future analyses by the Rutherford-Oxford-Lancs group, the uncertainties in the non-23 oscillation parameters *will* be taken into account in the confidence contours.

A comparison of the 90% 3-flavour confidence contours with previous results from MINOS [17] and Super-K [18] is also given using GENIE in figure 20 and using NEUT in figure 21.

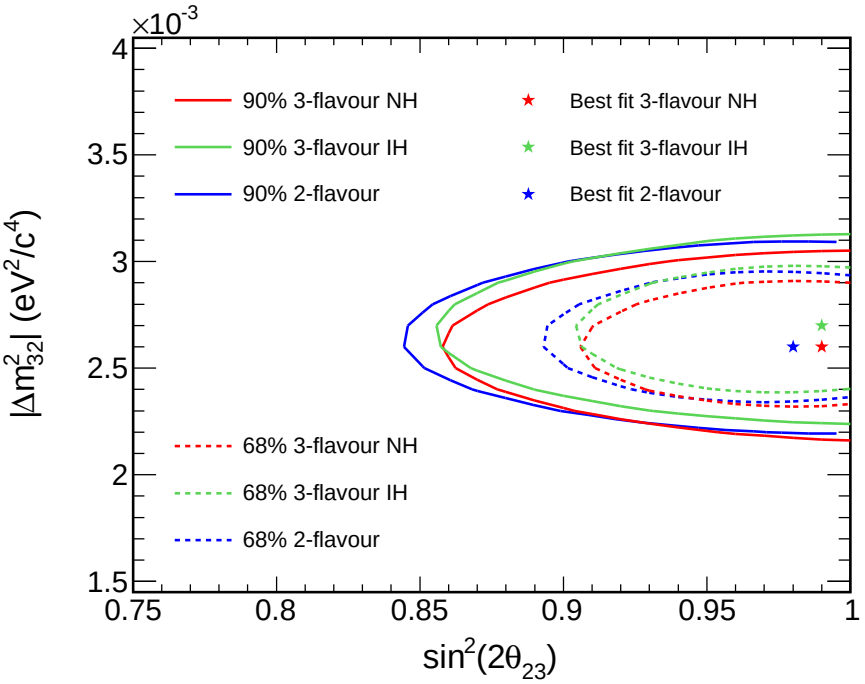


Figure 17: Comparison of confidence contours for $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ between 2-flavour, 3-flavour (NH) and 3-flavour (IH) using NEUT. All contours include both statistical and systematic uncertainties. The non-23 oscillation parameters are fixed.

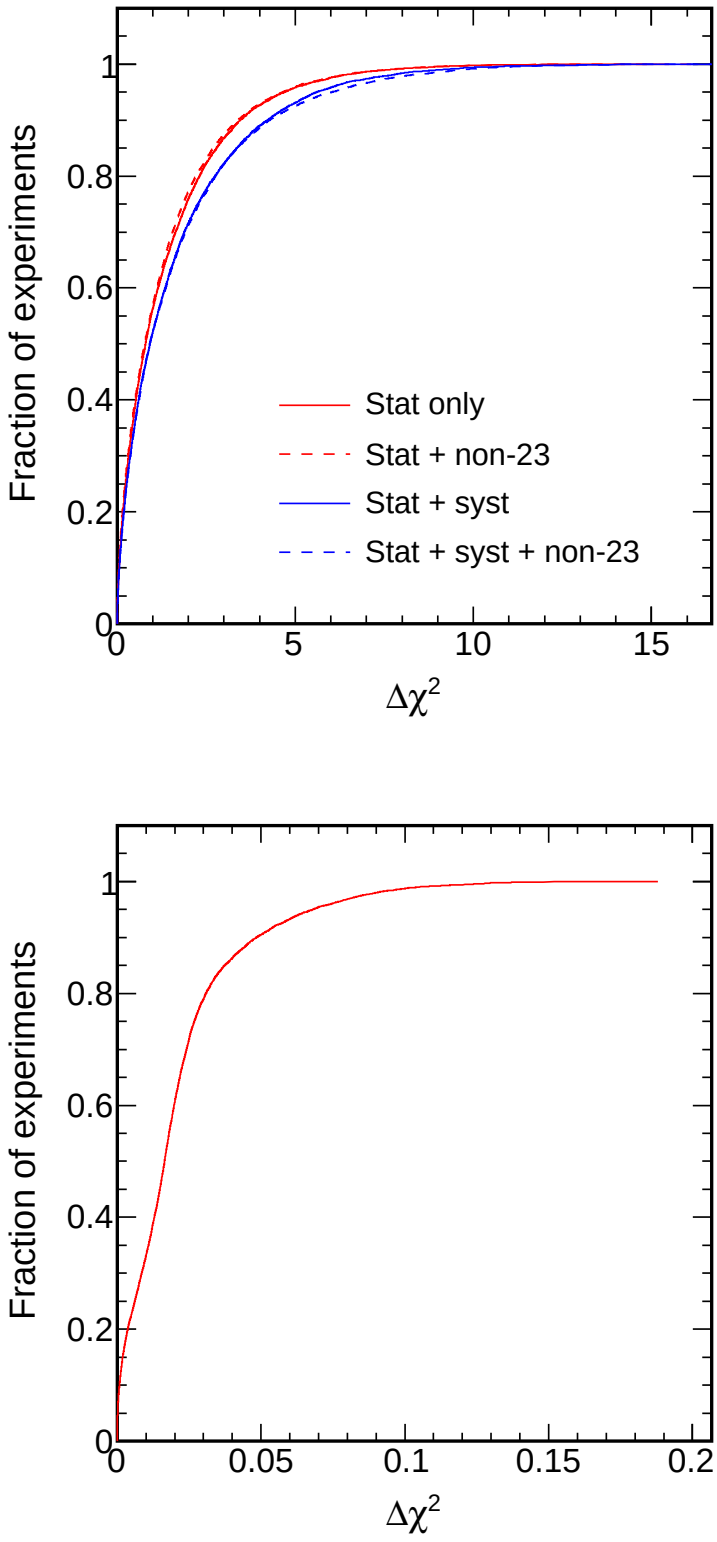


Figure 18: Cumulative distributions of $\Delta\chi^2$ (= χ^2_{TRUE} - $\chi^2_{BESTFIT}$) for 10000 toy MC experiments created at true values of $\sin^2(2\theta_{23})$ = 1.0 and $|\Delta m^2_{32}|$ = 3.1 $\times$ 10^{-3} eV^2/c^4 using NEUT; stat only, stat + non-23, stat + syst and stat + syst + non-23 (top) and non-23 only (bottom). Normal mass hierarchy, Earth crust density = 2.6 g/cm^3. When not varied, the non-23 oscillation parameters are fixed at $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, Δm^2_{21} = 7.6 $\times$ 10^{-5} eV^2/c^4, δ_{CP} = 0.

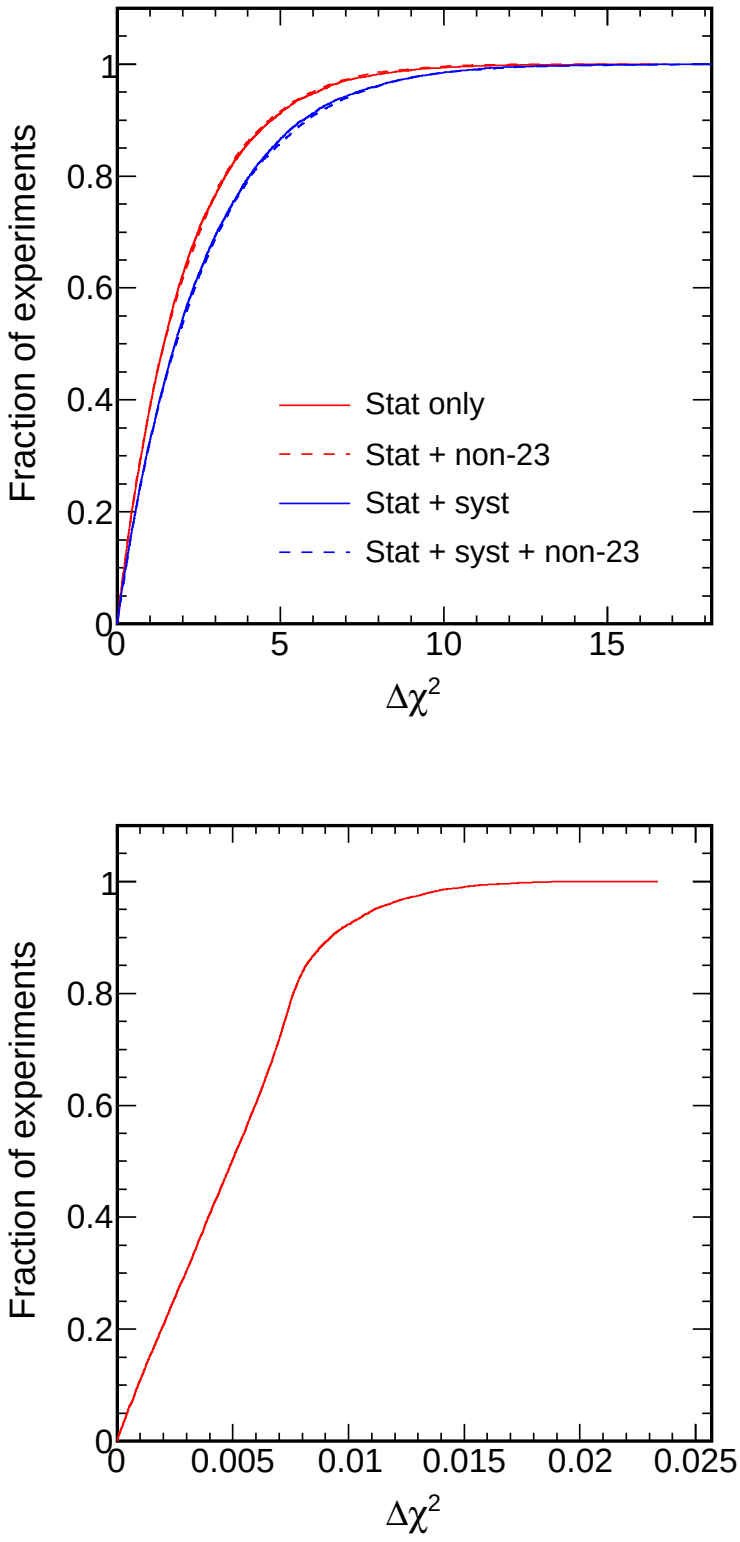


Figure 19: Cumulative distributions of $\Delta\chi^2$ (= χ^2_{TRUE} - $\chi^2_{BESTFIT}$) for 10000 toy MC experiments created at true values of $\sin^2(2\theta_{23})$ = 0.87 and $|\Delta m^2_{32}|$ = 2.4 $\times$ 10^{-3} eV^2/c^4 using NEUT; stat only, stat + non-23, stat + syst and stat + syst + non-23 (top) and non-23 only (bottom). Normal mass hierarchy, Earth crust density = 2.6 g/cm^3. When not varied, the non-23 oscillation parameters are fixed at $\sin^2(2\theta_{12})$ = 0.87, $\sin^2(2\theta_{13})$ = 0.1, Δm^2_{21} = 7.6 $\times$ 10^{-5} eV^2/c^4, δ_{CP} = 0.

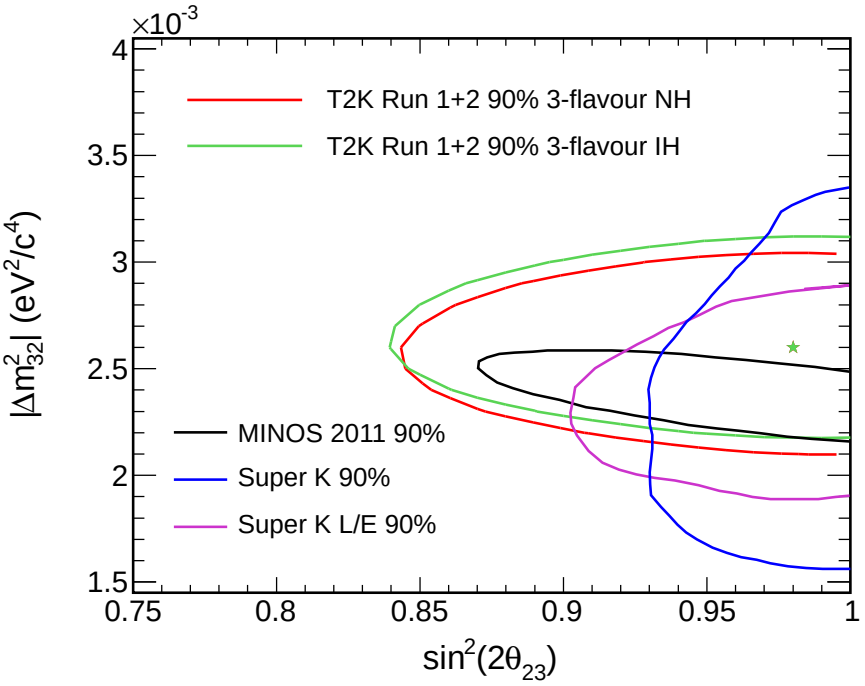


Figure 20: Comparison of T2K Run 1+2 3-flavour 90% confidence contours for $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ (GENIE) with results from MINOS [17] and Super-K [18]; the T2K 3-flavour best-fit points are identical for the NH and IH.

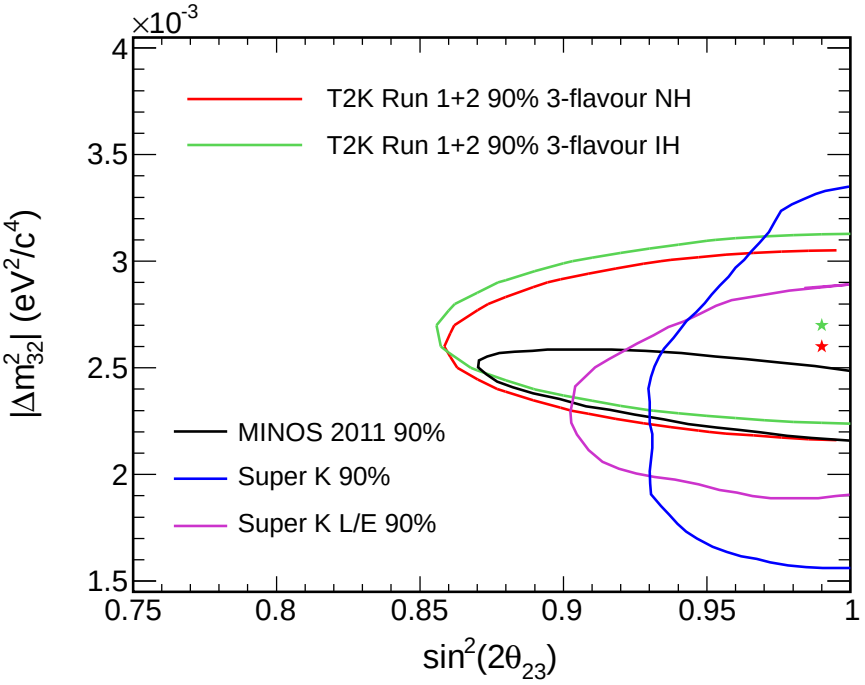


Figure 21: Comparison of T2K Run 1+2 3-flavour 90% confidence contours for $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ (NEUT) with results from MINOS [17] and Super-K [18]

6. Summary

In this note, we have presented the results of a 3-flavour ν_μ-disappearance analysis on the combined Run 1+2 dataset, which corresponds to an integrated J-PARC neutrino beam exposure of 1.431×10^{20} POT. The analysis method was identical to that used in our earlier analysis reported in Ref. [1] except that 3-flavour oscillation probabilities were used instead of the 2-flavour approximation and oscillated ν_e templates were added in the interaction modes CCQE, CC1π and CC other.

In the 3-flavour analysis, the total predicted number of single μ-like ring events in Super-K is higher for the inverted hierarchy (IH) than for the normal hierarchy (NH) when the atmospheric mass-squared difference is $|\Delta m^2_{32}|$. This is because more events are predicted above the oscillation maximum for the IH than for the NH and less are predicted below it. In turn this is due to the 3-flavour ν_μ-survival probability being higher for the IH than the NH above the oscillation maximum and lower below the oscillation maximum. Since there are more events above the oscillation maximum than below it, the total predicted number is higher for the IH than for the NH. This is true for any combination of the non-23 oscillation parameters.

Uncertainties in δ_{CP} can change the predicted reconstructed spectrum of single μ-like ring events in Super-K by up to 5% above the oscillation maximum and by up to 3% below it. This is larger than changes in the spectrum due to uncertainties in $\sin^2(2\theta_{13})$, $\sin^2(2\theta_{12})$ and Δm^2_{21} which are $\leq 0.5\%$.

A 3-flavour ν_μ-disappearance fit was made of the observed Super-K reconstructed energy spectrum of single μ-like ring events in the combined Run 1+2 dataset. In this fit, $\sin^2(2\theta_{23})$ and $|\Delta m^2_{32}|$ were varied to find their best-fit values, but the non-23 oscillation parameters were fixed. Using GENIE, this gave best-fit values of $\sin^2(2\theta_{23}) = 0.98$, $|\Delta m^2_{32}| = 2.6 \times 10^{-3}$ eV^2/c^4 for both mass hierarchies, while a fit using NEUT gave $\sin^2(2\theta_{23}) = 0.99$, $|\Delta m^2_{32}| = 2.6 \times 10^{-3}$ eV^2/c^4 for the NH and $\sin^2(2\theta_{23}) = 0.99$, $|\Delta m^2_{32}| = 2.7 \times 10^{-3}$ eV^2/c^4 for the IH. For comparison, the best-fit values from the 2-flavour analysis in [1] were $\sin^2(2\theta_{23}) = 0.97$, $|\Delta m^2_{32}| = 2.6 \times 10^{-3}$ eV^2/c^4 using GENIE and $\sin^2(2\theta_{23}) = 0.98$, $|\Delta m^2_{32}| = 2.6 \times 10^{-3}$ eV^2/c^4 using NEUT. The 3-flavour best-fit points were higher in $\sin^2(2\theta_{23})$ than those in the 2-flavour analysis due to the effect of $\sin^2(2\theta_{13})$: in the 3-flavour analysis $\sin^2(2\theta_{13}) = 0.1$, whereas in the 2-flavour analysis the approximation $\sin^2(2\theta_{13}) = 0$ was used.

Uncertainties in the best-fit points due to uncertainties in δ_{CP} are larger than those due to uncertainties in $\sin^2(2\theta_{13})$, $\sin^2(2\theta_{12})$ and Δm^2_{21}; these uncertainties due to δ_{CP} are 0.5% (NH) and 0.35% (IH) in $\sin^2(2\theta_{23})$ and 0.6% (NH) and 0.4% (IH) in $|\Delta m^2_{32}|$. However this is relatively small compared with the combined systematic uncertainties in the best-fit points which are 4.3 - 5.1%.

The 3-flavour confidence contours are narrower in $\sin^2(2\theta_{23})$ than the 2-flavour contours since the 3-flavour best-fit points are nearer to $\sin^2(2\theta_{23}) = 1.0$. The confidence contours for the IH include higher values of $|\Delta m^2_{32}|$ than those for the NH; this is because it is necessary to choose a higher value of this parameter for the IH than for the NH in order to fit the oscillation dip at a certain energy in the real data.

Acknowledgement

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